

## CHAPTER 6

**6-1 OVERVIEW** So far we have only dealt with fields that do not change with time. In constructing the electrostatic model we defined an electric field intensity vector,  $\mathbf{E}$ , and an electric flux density (electric displacement) vector,  $\mathbf{D}$ . The fundamental governing differential equations are

$$\nabla \times \mathbf{E} = 0, \quad (3-4)(6-1)$$

$$\nabla \cdot \mathbf{D} = \rho_v. \quad (3-63)(6-2)$$

For linear and isotropic (not necessarily homogeneous) media,  $\mathbf{E}$  and  $\mathbf{D}$  are related by the constitutive relation

$$\mathbf{D} = \epsilon \mathbf{E}. \quad (3-67)(6-3)$$

For the magnetostatic model we defined a magnetic flux density vector,  $\mathbf{B}$ , and a magnetic field intensity vector,  $\mathbf{H}$ . The fundamental governing differential equations are

$$\nabla \cdot \mathbf{B} = 0, \quad (5-6)(6-4)$$

$$\nabla \times \mathbf{H} = \mathbf{J}. \quad (5-61)(6-5)$$

The constitutive relation for  $\mathbf{B}$  and  $\mathbf{H}$  in linear and isotropic media is

$$\mathbf{H} = \frac{1}{\mu} \mathbf{B}. \quad (5-66)(6-6)$$

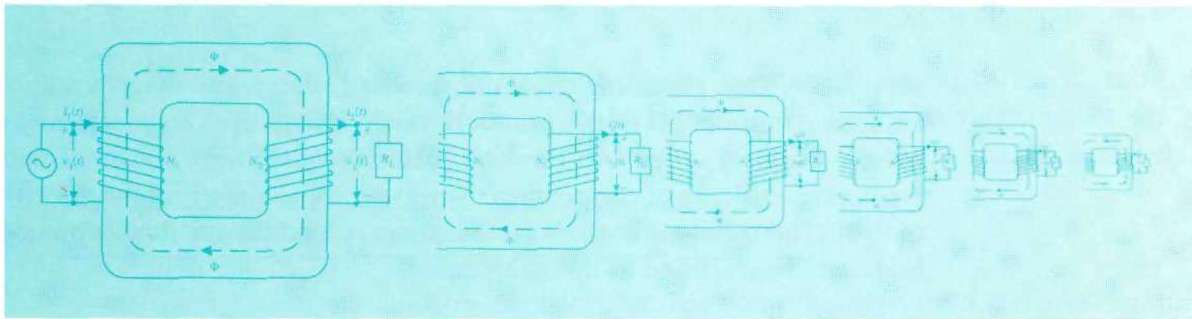
We observe that  $\mathbf{E}$  and  $\mathbf{D}$  in the electrostatic model are not related to  $\mathbf{B}$  and  $\mathbf{H}$  in the magnetostatic model. In a conducting medium, static electric and

Fundamental governing equations for electrostatic model

Constitutive relation for the electric model

Fundamental governing equations for magnetostatic model

Constitutive relation for the magnetic model



# Time-Varying Fields and Maxwell's Equations

## Electromagnetostatic field

magnetic fields may both exist and form an *electromagnetostatic field*. A static electric field in a conducting medium causes a steady current to flow that, in turn, gives rise to a static magnetic field. However, the electric field can be completely determined from the static electric charges or potential distributions. The magnetic field is a consequence; it does not enter into the calculation of the electric field.

Static models are simple, but they are inadequate for explaining time-varying electromagnetic phenomena. Static electric and magnetic fields do not give rise to waves that propagate and carry energy and information. Waves are the essence of electromagnetic action at a distance. In this chapter we will see that a changing magnetic field induces an electric field, and vice versa. Under time-varying conditions it is necessary to construct an electromagnetic model in which the electric field vectors **E** and **D** are properly related to the magnetic field vectors **B** and **H**.

We will begin with a fundamental postulate that modifies the  $\nabla \times \mathbf{E}$  equation in Eq. (6-1) and leads to Faraday's law of electromagnetic induction. The concepts of transformer emf and motional emf will be discussed. With the new postulate we will also need to modify the  $\nabla \times \mathbf{H}$  equation in order to make the governing equations consistent with the law of conservation of charge. The two modified curl equations together with the two divergence equations (6-2) and (6-4), are known as Maxwell's equations and form the foundation of electromag-

netic theory. The governing equations for electrostatics and magnetostatics are special forms of Maxwell's equations when all quantities are independent of time. Maxwell's equations can be combined to yield wave equations that predict the existence of electromagnetic waves propagating with the velocity of light. The solutions of the wave equations, especially for time-harmonic fields, will be discussed in this chapter.

## 6-2 FARADAY'S LAW OF ELECTROMAGNETIC INDUCTION

A major advance in electromagnetic theory was made by Michael Faraday, who, in 1831, discovered experimentally that a current was induced in a conducting loop when the magnetic flux linking the loop changed. The quantitative relationship between the induced emf and the rate of change of flux linkage, based on experimental observation, is known as *Faraday's law*. It is an experimental law and can be considered as a postulate. However, we do not take the experimental relation concerning a finite loop as the starting point for developing the theory of electromagnetic induction. Instead, we follow our approach in Chapter 3 for electrostatics and in Chapter 5 for magnetostatics by stating a fundamental postulate and developing from it the integral forms of Faraday's law.

*The fundamental postulate for electromagnetic induction is*

Fundamental  
postulate for  
electromagnetic  
induction

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t} \quad (6-7)$$

Equation (6-7) expresses a point-function relationship; that is, it applies to every point in space, whether it be in free space or in a material medium. *The electric field intensity in a region of time-varying magnetic flux density is therefore nonconservative and cannot be expressed as the negative gradient of a scalar potential.*

Electric field  
intensity in a time-  
varying magnetic  
field is not  
conservative.

Taking the surface integral of both sides of Eq. (6-7) over an open surface and applying Stokes's theorem, we obtain

$$\oint_C \mathbf{E} \cdot d\boldsymbol{\ell} = - \int_S \frac{\partial \mathbf{B}}{\partial t} \cdot d\mathbf{s} \quad (6-8)$$

Equation (6-8) is valid for any surface  $S$  with a bounding contour  $C$ , whether or not a physical circuit exists around  $C$ . Of course, in a field with no time variation,  $\partial \mathbf{B} / \partial t = 0$ , Eqs. (6-7) and (6-8) reduce to Eqs. (6-1) and (3-7), respectively, for electrostatics.



In the following subsections we discuss separately the cases of a stationary circuit in a time-varying magnetic field, a moving conductor in a static magnetic field, and a moving circuit in a time-varying magnetic field.

### 6-2.1 A STATIONARY CIRCUIT IN A TIME-VARYING MAGNETIC FIELD

For a stationary circuit with a contour  $C$  and surface  $S$ , Eq. (6-8) can be written as

$$\oint_C \mathbf{E} \cdot d\boldsymbol{\ell} = -\frac{d}{dt} \int_S \mathbf{B} \cdot d\mathbf{s}. \quad (6-9)$$

If we define

$$\mathcal{V} = \oint_C \mathbf{E} \cdot d\boldsymbol{\ell} = \text{emf induced in circuit with contour } C \quad (\text{V}) \quad (6-10)$$

and

$$\Phi = \int_S \mathbf{B} \cdot d\mathbf{s} = \text{magnetic flux crossing surface } S \quad (\text{Wb}), \quad (6-11)$$

then Eq. (6-9) becomes

$$\boxed{\mathcal{V} = -\frac{d\Phi}{dt} \quad (\text{V}).} \quad (6-12)$$

Equation (6-12) states that *the electromotive force induced in a stationary closed circuit is equal to the negative rate of increase of the magnetic flux linking the circuit*. This is a statement of *Faraday's law of electromagnetic induction*. The negative sign in Eq. (6-12) is an assertion that the induced emf will cause a current to flow in the closed loop in such a direction as to oppose the change in the linking magnetic flux. This assertion is known as *Lenz's law*. The emf induced in a stationary loop caused by a time-varying magnetic field is a *transformer emf*.

Faraday's law of electromagnetic induction

Lenz's law of transformer emf

### EXAMPLE 6-1

A circular loop of  $N$  turns of conducting wire lies in the  $xy$ -plane with its center at the origin of a magnetic field specified by  $\mathbf{B} = \mathbf{a}_z B_0 \cos(\pi r/2b) \sin \omega t$ , where  $b$  is the radius of the loop and  $\omega$  is the angular frequency. Find the emf induced in the loop.

### SOLUTION

The problem specified a stationary loop in a time-varying magnetic field; hence Eq. (6-12) can be used directly to find the induced emf,  $\mathcal{V}$ . The



magnetic flux linking each turn of the circular loop is

$$\begin{aligned}\Phi &= \int_S \mathbf{B} \cdot d\mathbf{s} \\ &= \int_0^b \left[ \mathbf{a}_z B_0 \cos \left( \frac{\pi r}{2b} \right) \sin \omega t \right] \cdot (\mathbf{a}_z 2\pi r dr) \\ &= \frac{8b^2}{\pi} \left( \frac{\pi}{2} - 1 \right) B_0 \sin \omega t.\end{aligned}$$

Since there are  $N$  turns, the total flux linkage is  $N\Phi$ , and we obtain

$$\begin{aligned}\mathcal{V} &= -N \frac{d\Phi}{dt} \\ &= -\frac{8N}{\pi} b^2 \left( \frac{\pi}{2} - 1 \right) B_0 \omega \cos \omega t \quad (\text{V}).\end{aligned}$$

As the phase of  $\cos \omega t$  leads that of  $\sin \omega t$  by  $90^\circ$ , we see that the phase of the induced emf lags behind that of the flux linkage by  $90^\circ$ .

### ■ EXERCISE 6.1

Find the emf induced in the  $N$ -turn circular loop of radius  $b$  in Example 6-1 if the magnetic flux density is  $\mathbf{B} = \mathbf{a}_z B_0 (b-r) \cos \omega t$ . What is the phase relationship between the induced emf and the magnetic field?

ANS.  $\mathcal{V} = \frac{\pi}{3} \omega b^3 B_0 \sin \omega t$ , lags  $90^\circ$  behind  $\mathbf{B}$  in phase.

## 6-2.2 TRANSFORMERS

### Functions of a transformer

A transformer is an alternating-current (a-c) device that transforms voltages, currents, and impedances. It usually consists of two or more coils coupled magnetically through a common ferromagnetic core, such as that sketched in Fig. 6-1. For the closed path in the magnetic circuit traced by magnetic flux  $\Phi$ , we have

$$N_1 i_1 - N_2 i_2 = \mathcal{R} \Phi, \quad (6-13)$$

where  $N_1$ ,  $N_2$  and  $i_1$ ,  $i_2$  are the numbers of turns and the currents in the primary and secondary circuits, respectively. The left side of Eq. (6-13) is the closed line integral  $\oint \mathbf{H} \cdot d\mathbf{l}$  around the transformer core, a consequence of Ampère's circuital law as expressed in Eq. (5-63), and represents the net magnetomotive force (mmf—SI unit: ampere-turn). We have noted, in accordance with Lenz's law, that the induced mmf in the secondary circuit,  $N_2 i_2$ , opposes the flow of magnetic flux created by the mmf in the primary circuit,  $N_1 i_1$ . On the right side of Eq. (6-13)  $\mathcal{R}$  denotes the *reluctance* of the

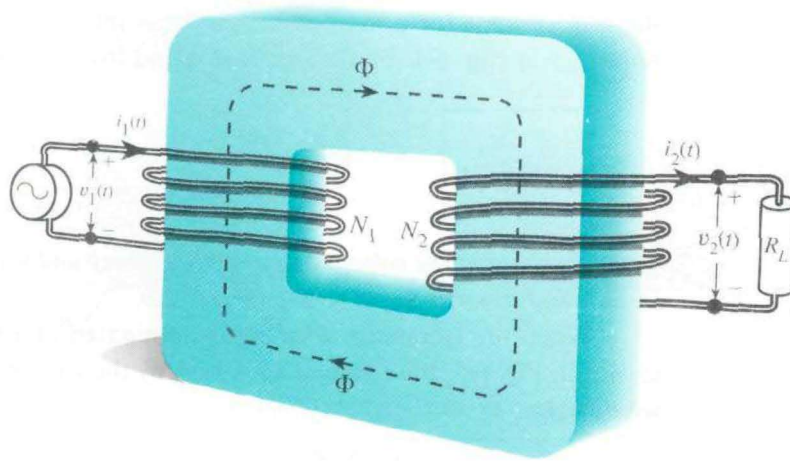


FIGURE 6-1 Schematic diagram of a transformer.

magnetic circuit, which depends on the geometry and is inversely proportional to the permeability of the core material.<sup>†</sup> Equation (6-13) for a magnetic circuit is analogous to an expression of Kirchhoff's voltage law for a d-c electric circuit, which says that the net emf around a closed circuit is equal to the sum of the resistances times the current. Here  $\mathcal{R}$  and  $\Phi$  are analogous to resistance and current respectively.

For ideal transformers we assume there is no leakage flux, and that  $\mu \rightarrow \infty$ ,  $\mathcal{R} = 0$ . Equation (6-13) becomes

$$\frac{i_1}{i_2} = \frac{N_2}{N_1}. \quad (6-14)$$

Equation (6-14) states that *the ratio of the currents in the primary and secondary windings of an ideal transformer is equal to the inverse ratio of the numbers of turns*. Faraday's law tells us that

$$v_1 = N_1 \frac{d\Phi}{dt} \quad (6-15)$$

and

$$v_2 = N_2 \frac{d\Phi}{dt}, \quad (6-16)$$

<sup>†</sup>The calculation of the reluctance of a magnetic circuit is similar to that of a resistance in an electric circuit, but the result can only be approximate because of the existence of leakage flux and the nonuniformity of the cross-sectional area of the transformer core. (Leakage flux is that part of magnetic flux that does not link with the entire magnetic circuit.) We will not elaborate on the determination of  $\mathcal{R}$  here. The SI unit for reluctance is reciprocal henry ( $\text{H}^{-1}$ ).

Conditions of an  
ideal transformer

Current relation for  
an ideal transformer

the proper signs for  $v_1$  and  $v_2$  having been taken care of by the designated polarities in Fig. 6-1. From Eqs. (6-15) and (6-16) we have

$$\frac{v_1}{v_2} = \frac{N_1}{N_2}. \quad (6-17)$$

Voltage relation for  
an ideal transformer

Thus, *the ratio of the voltages across the primary and secondary windings of an ideal transformer is equal to the turns ratio.*

When the secondary winding is terminated in a load resistance  $R_L$ , as shown in Fig. 6-1, the effective load seen by the source connected to primary winding is

$$(R_1)_{\text{eff}} = \frac{v_1}{i_1} = \frac{(N_1/N_2)v_2}{(N_2/N_1)i_2}, \quad (6-18)$$

or

Resistance  
transformation by an  
ideal transformer

$$(R_1)_{\text{eff}} = \left(\frac{N_1}{N_2}\right)^2 R_L, \quad (6-19)$$

which is the load resistance multiplied by the square of the turns ratio. For a sinusoidal source  $v_1(t)$  and a load impedance  $Z_L$ , the effective load seen by the source is  $(N_1/N_2)^2 Z_L$ , an impedance transformation. We have

Impedance  
transformation by an  
ideal transformer

$$(Z_1)_{\text{eff}} = \left(\frac{N_1}{N_2}\right)^2 Z_L. \quad (6-20)$$

We note from Section 5-10 that the inductance of a coil is proportional to the permeability of the medium. Thus, the assumption of an infinite  $\mu$  for an ideal transformer also implies infinite inductances.

Why an exact  
analysis of a real  
transformer is  
difficult

For real transformers we have the following real-life conditions: the existence of leakage flux, noninfinite inductances, nonzero winding resistances, and the presence of hysteresis and eddy-current losses. The nonlinear nature of the ferromagnetic core (the dependence of permeability on magnetic field intensity) further compounds the difficulty of an exact analysis of real transformers.

Definition of eddy  
current

When time-varying magnetic flux flows in the ferromagnetic core, an induced emf will result in accordance with Faraday's law. This induced emf will produce local currents in the conducting core normal to the magnetic flux. These currents are called **eddy currents**. Eddy currents produce ohmic power loss and cause local heating. As a matter of fact, this is the principle of induction heating. Induction furnaces have been built to produce high

Principle of  
induction heating



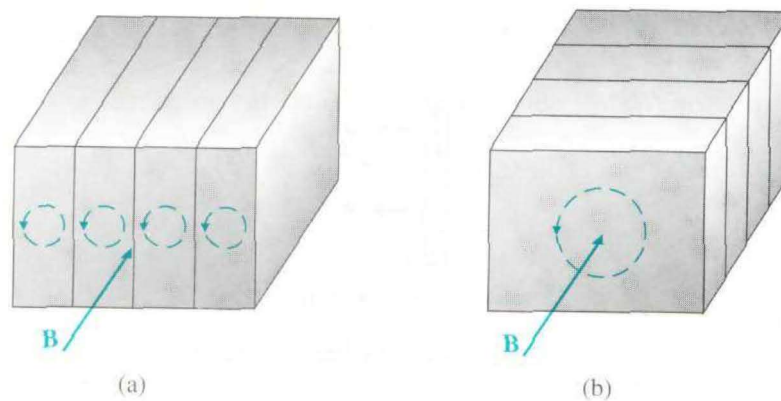


FIGURE 6-2 Laminated cores in a time-varying magnetic field.

Core lamination for  
reducing eddy-  
current power loss

enough temperatures to melt metals. In transformers this eddy-current power loss is undesirable and can be reduced by using core materials that have high permeability but low conductivity (high  $\mu$  and low  $\sigma$ ). Ferrites are such materials. For low-frequency, high-power applications, an economical way for reducing eddy-current power loss is to use laminated cores; that is, to make transformer cores out of stacked ferromagnetic (iron) sheets, each electrically insulated from its neighbors by thin varnish or oxide coatings. The insulating coatings should be parallel to the direction of the magnetic flux, as shown in Fig. 6-2(a), so that eddy currents normal to the flux are restricted to the laminated sheets. The arrangement in Fig. 6-2(b) with insulated layers normal to the magnetic flux clearly has little effect in reducing eddy-current power loss. It can be proved that the total eddy-current power loss decreases as the number of laminations increases. The amount of power-loss reduction depends on the shape and size of the cross section as well as on the method of lamination.

### EXERCISE 6.2

A resistance of  $75(\Omega)$  is to be converted into  $300(\Omega)$  by means of an ideal transformer. What turns-ratio should the transformer have?

ANS. 2:1.

### 6-2.3 A MOVING CONDUCTOR IN A STATIC MAGNETIC FIELD

When a conductor moves with a velocity  $\mathbf{u}$  in a static (non-time-varying) magnetic field  $\mathbf{B}$  as shown in Fig. 6-3, a force  $\mathbf{F}_m = q\mathbf{u} \times \mathbf{B}$  will cause the freely movable electrons in the conductor to drift toward one end of the conductor and leave the other end positively charged. This separation of the positive and negative charges creates a Coulombian force of attraction. The charge-separation process continues until the electric and magnetic forces balance each other and a state of equilibrium is reached. At equilibrium,

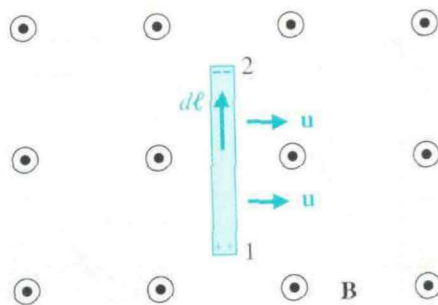


FIGURE 6-3 A conducting bar moving in a magnetic field.

which is reached very rapidly, the net force on the free charges in the moving conductor is zero.

To an observer moving with the conductor there is no apparent motion, and the magnetic force per unit charge  $\mathbf{F}_m/q = \mathbf{u} \times \mathbf{B}$  can be interpreted as an induced electric field acting along the conductor and producing a voltage

$$V_{21} = \int_1^2 (\mathbf{u} \times \mathbf{B}) \cdot d\boldsymbol{\ell}. \quad (6-21)$$

If the moving conductor is a part of a closed circuit  $C$ , then the emf generated around the circuit is

$$\mathcal{V}' = \oint_C (\mathbf{u} \times \mathbf{B}) \cdot d\boldsymbol{\ell} \quad (\text{V}). \quad (6-22)$$

Definition of  
motional emf

This is referred to as a **flux-cutting emf** or a **motional emf**. Obviously, only the part of the circuit that moves in a direction not parallel to (and hence, figuratively, “cutting”) the magnetic flux will contribute to  $\mathcal{V}'$  in Eq. (6-22).

### EXAMPLE 6-2

A metal bar slides over a pair of conducting rails in a uniform magnetic field  $\mathbf{B} = \mathbf{a}_z B_0$  with a constant velocity  $\mathbf{u}$ , as shown in Fig. 6-4.

- Determine the open-circuit voltage  $V_0$  that appears across terminals 1 and 2.
- Assuming that a resistance  $R$  is connected between the terminals, find the electric power dissipated in  $R$ .
- Show that this electric power is equal to the mechanical power required to move the sliding bar with a velocity  $\mathbf{u}$ . Neglect the electric resistance of the metal bar and of the conducting rails. Neglect also the mechanical friction at the contact points.

## SOLUTION

- a) The moving bar generates a flux-cutting emf. We use Eq. (6-22) to find the open-circuit voltage  $V_0$ :

$$\begin{aligned} V_0 = V_1 - V_2 &= \oint_C (\mathbf{u} \times \mathbf{B}) \cdot d\boldsymbol{\ell} \\ &= \int_{2'}^{1'} (\mathbf{a}_x u \times \mathbf{a}_z B_0) \cdot (\mathbf{a}_y d\ell) \\ &= -uB_0h \quad (\text{V}). \end{aligned} \quad (6-23)$$

- b) When a resistance  $R$  is connected between terminals 1 and 2, a current  $I = uB_0h/R$  will flow from terminal 2 to terminal 1, so the electric power,  $P_e$ , dissipated in  $R$  is

$$P_e = I^2 R = \frac{(uB_0h)^2}{R} \quad (\text{W}). \quad (6-24)$$

- c) The mechanical power,  $P_m$ , required to move the sliding bar is

$$P_m = \mathbf{F} \cdot \mathbf{u} \quad (\text{W}), \quad (6-25)$$

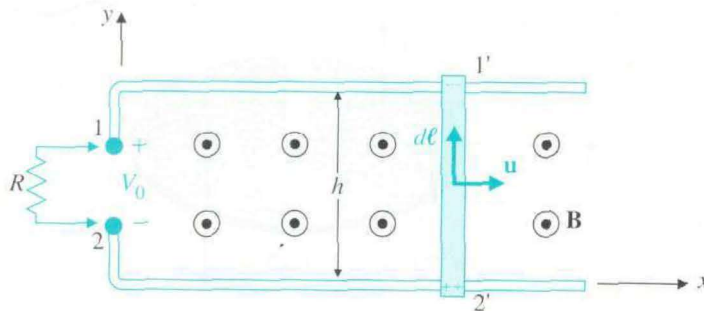
where  $\mathbf{F}$  is the mechanical force required to counteract the magnetic force,  $\mathbf{F}_m$ , which the magnetic field exerts on the current-carrying metal bar. From Eq. (5-116) we have

$$\mathbf{F}_m = I \int_{2'}^{1'} d\boldsymbol{\ell} \times \mathbf{B} = -\mathbf{a}_x IB_0h \quad (\text{N}). \quad (6-26)$$

The negative sign in Eq. (6-26) arises because current  $I$  flows in a direction opposite to that of  $d\boldsymbol{\ell}$ . Hence,

$$\mathbf{F} = -\mathbf{F}_m = \mathbf{a}_x IB_0h = \mathbf{a}_x uB_0^2h^2/R \quad (\text{N}). \quad (6-27)$$

FIGURE 6-4 A metal bar sliding over conducting rails (Example 6-2).





Substitution of Eq. (6-27) in Eq. (6-25) proves that  $P_m = P_e$ , which upholds the principle of conservation of energy.

### EXAMPLE 6-3

Faraday disk  
generator

The **Faraday disk generator** consists of a circular metal disk rotating with a constant angular velocity  $\omega$  in a uniform and constant magnetic field of flux density  $\mathbf{B} = \mathbf{a}_z B_0$  that is parallel to the axis of rotation. Brush contacts are provided at the axis and on the rim of the disk, as depicted in Fig. 6-5. Determine the open-circuit voltage of the generator if the radius of the disk is  $b$ .

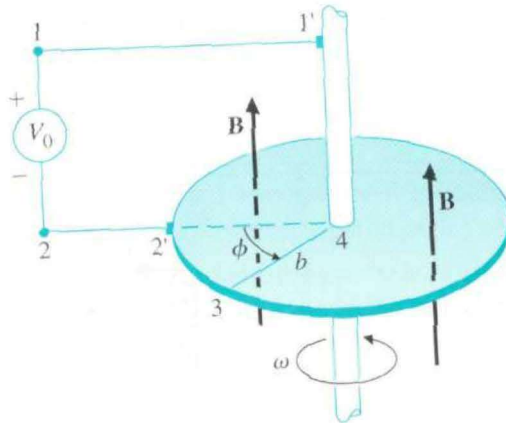
### SOLUTION

Let us consider the circuit 122'341'. Of the part 2'34 that moves with the disk, only the straight portion 34 "cuts" the magnetic flux. We have, from Eq. (6-22),

$$\begin{aligned} V_0 &= \oint (\mathbf{u} \times \mathbf{B}) \cdot d\ell \\ &= \int_3^4 [(\mathbf{a}_\phi r \omega) \times \mathbf{a}_z B_0] \cdot (\mathbf{a}_r dr) \\ &= \omega B_0 \int_b^0 r dr = -\frac{\omega B_0 b^2}{2} \quad (\text{V}), \end{aligned} \quad (6-28)$$

which is the emf of the Faraday disk generator. To measure  $V_0$ , we must use a voltmeter of a very high resistance so that no appreciable current flows in the circuit to modify the externally applied magnetic field.

FIGURE 6-5 Faraday disk generator (Example 6-3).



## 6-2.4 A MOVING CIRCUIT IN A TIME-VARYING MAGNETIC FIELD

When a charge  $q$  moves with a velocity  $\mathbf{u}$  in a region where both an electric field  $\mathbf{E}$  and a magnetic field  $\mathbf{B}$  exist, the electromagnetic force  $\mathbf{F}$  on  $q$ , as measured by a laboratory observer, is given by Lorentz's force equation:

$$\mathbf{F} = q(\mathbf{E} + \mathbf{u} \times \mathbf{B}). \quad (5-5)(6-29)$$

Lorentz's force  
equation

To an observer moving with  $q$ , there is no apparent motion, and the force on  $q$  can be interpreted as caused by an electric field  $\mathbf{E}'$ , where

$$\mathbf{E}' = \mathbf{E} + \mathbf{u} \times \mathbf{B}, \quad (6-30)$$

or

$$\mathbf{E} = \mathbf{E}' - \mathbf{u} \times \mathbf{B}. \quad (6-31)$$

Hence, when a conducting circuit with contour  $C$  and surface  $S$  moves with a velocity  $\mathbf{u}$  in a field  $(\mathbf{E}, \mathbf{B})$ , we use Eq. (6-31) in Eq. (6-8) to obtain

General form of  
Faraday's law

$$\oint_C \mathbf{E}' \cdot d\boldsymbol{\ell} = - \int_S \frac{\partial \mathbf{B}}{\partial t} \cdot d\mathbf{s} + \oint_C (\mathbf{u} \times \mathbf{B}) \cdot d\boldsymbol{\ell} \quad (\text{V}). \quad (6-32)$$

Equation (6-32) is the general form of *Faraday's law* for a moving circuit in a time-varying magnetic field. The line integral on the left side is the emf induced in the moving frame of reference. The first term on the right side represents the transformer emf due to the time variation of  $\mathbf{B}$ , and the second term represents the motional emf due to the motion of the circuit in  $\mathbf{B}$ .

If we designate the left side of Eq. (6-32) by

$$\begin{aligned} \mathcal{V}' &= \oint_C \mathbf{E}' \cdot d\boldsymbol{\ell} \\ &= \text{emf induced in circuit } C \text{ measured in the moving frame,} \end{aligned} \quad (6-33)$$

it can be proved in general that Eq. (6-32) is equivalent to

Another form of  
Faraday's law

$$\begin{aligned} \mathcal{V}' &= - \frac{d}{dt} \int_S \mathbf{B} \cdot d\mathbf{s} \\ &= - \frac{d\Phi}{dt} \quad (\text{V}), \end{aligned} \quad (6-34)$$

which is of the same form as Eq. (6-12). Of course, if a circuit is not in motion,  $\mathcal{V}'$  reduces to  $\mathcal{V}$ . Hence, Faraday's law that the emf induced in a closed circuit equals the negative time-rate of increase of the magnetic flux linking a

circuit applies to a stationary circuit as well as a moving one. Either Eq. (6-32) or Eq. (6-34) can be used to evaluate the induced emf in the general case.

#### EXAMPLE 6-4

Determine the open-circuit voltage of the Faraday disk generator in Example 6-3 by using Eq. (6-34).

#### SOLUTION

We solved the problem in Example 6-3 by using Eq. (6-22), which is Eq. (6-32) with  $\partial \mathbf{B} / \partial t = 0$ . In order to use Eq. (6-34), we first find the magnetic flux linking the circuit 122'341'1 in Fig. 6-5, which is the flux passing through the wedge-shaped area 2'342'.

$$\begin{aligned}\Phi &= \int_S \mathbf{B} \cdot d\mathbf{s} = B_0 \int_0^b \int_0^{\omega t} r \, d\phi \, dr \\ &= B_0(\omega t) \frac{b^2}{2}\end{aligned}\quad (6-35)$$

and

$$V_0 = \mathcal{E} = -\frac{d\Phi}{dt} = -\frac{\omega B_0 b^2}{2}, \quad (6-36)$$

which is the same as Eq. (6-28).

#### EXERCISE 6.3

Determine the open-circuit voltage that appears across terminals 1 and 2 in Example 6-2 by using Eq. (6-34).

ANS.  $-uB_0h$ .

#### EXAMPLE 6-5

An  $h$  by  $w$  rectangular conducting loop is situated in a changing magnetic field  $\mathbf{B} = \mathbf{a}_y B_0 \sin \omega t$ . The normal of the loop initially makes an angle  $\alpha$  with  $\mathbf{a}_y$ , as shown in Fig. 6-6. Find the induced emf in the loop: (a) when the loop is at rest, and (b) when the loop rotates with an angular velocity  $\omega$  about the  $x$ -axis.

#### SOLUTION

a) When the loop is at rest, we use Eq. (6-12):

$$\begin{aligned}\Phi &= \int \mathbf{B} \cdot d\mathbf{s} \\ &= (\mathbf{a}_y B_0 \sin \omega t) \cdot (\mathbf{a}_n hw) \\ &= B_0 hw \sin \omega t \cos \alpha.\end{aligned}$$

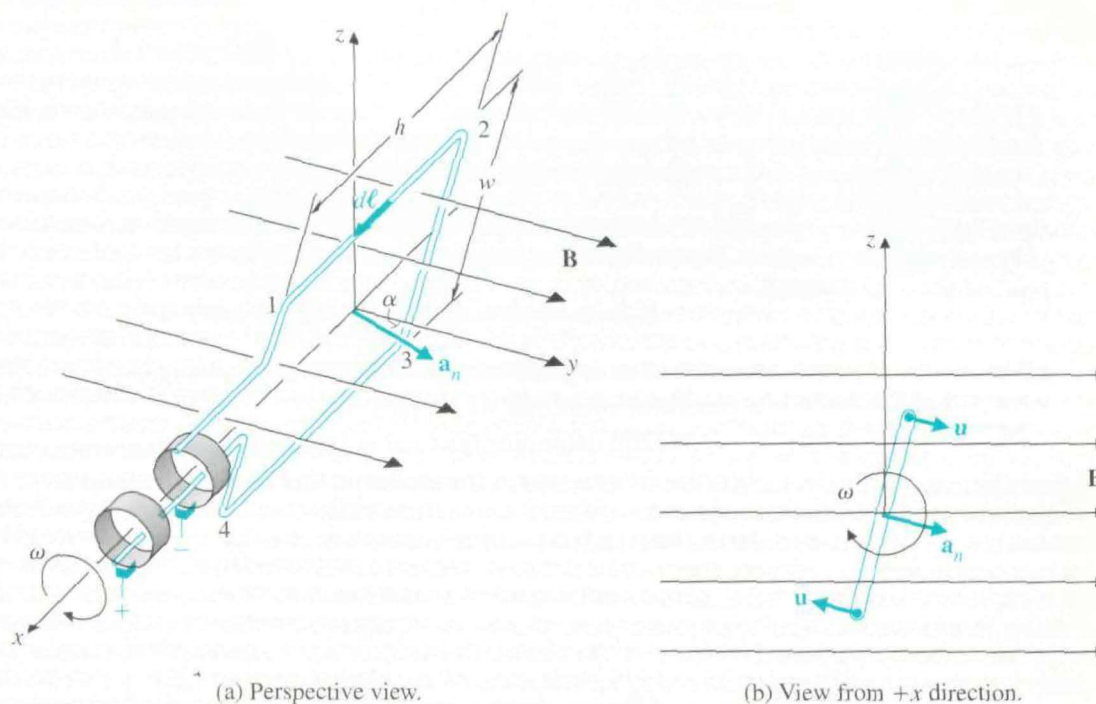


FIGURE 6-6 A rectangular conducting loop rotating in a changing magnetic field (Example 6-5).

Therefore,

$$\mathcal{V}_a = -\frac{d\Phi}{dt} = -B_0 S \omega \cos \omega t \cos \alpha, \quad (6-37)$$

where  $S = hw$  is the area of the loop. The relative polarities of the terminals are as indicated. If the circuit is completed through an external load,  $\mathcal{V}_a$  will produce a current that will oppose the change in  $\Phi$ .

- b) When the loop rotates about the  $x$ -axis, both terms in Eq. (6-32) contribute: the first term contributes the transformer emf  $\mathcal{V}_a$  in Eq. (6-37), and the second term contributes a motional emf  $\mathcal{V}_a'$ , where

$$\begin{aligned} \mathcal{V}_a' &= \oint_C (\mathbf{u} \times \mathbf{B}) \cdot d\boldsymbol{\ell} \\ &= \int_2^1 \left[ \left( \mathbf{a}_n \frac{w}{2} \omega \right) \times (\mathbf{a}_y B_0 \sin \omega t) \right] \cdot (\mathbf{a}_x dx) \\ &\quad + \int_4^3 \left[ \left( -\mathbf{a}_n \frac{w}{2} \omega \right) \times (\mathbf{a}_y B_0 \sin \omega t) \right] \cdot (\mathbf{a}_x dx) \\ &= 2 \left( \frac{w}{2} \omega B_0 \sin \omega t \sin \alpha \right) h. \end{aligned}$$



Note that the sides 23 and 41 do not contribute to  $\mathcal{V}_a'$  and that the contributions of sides 12 and 34 are of equal magnitude and in the same direction. If  $\alpha = 0$  at  $t = 0$ , then  $\alpha = \omega t$ , and we can write

$$\mathcal{V}_a' = B_0 S \omega \sin \omega t \sin \omega t. \quad (6-38)$$

The total emf induced or generated in the rotating loop is the sum of  $\mathcal{V}_a$  in Eq. (6-37) and  $\mathcal{V}_a'$  in Eq. (6-38):

$$\mathcal{V}_t' = -B_0 S \omega (\cos^2 \omega t - \sin^2 \omega t) = -B_0 S \omega \cos 2\omega t, \quad (6-39)$$

which has an angular frequency  $2\omega$ . Thus the arrangement in Fig. 6-6 is a generator of second harmonics.

We can determine the total induced emf  $\mathcal{V}_t'$  by applying Eq. (6-34) directly. At any time  $t$ , the magnetic flux linking the loop is

$$\begin{aligned} \Phi(t) &= \mathbf{B}(t) \cdot [\mathbf{a}_n(t)S] = B_0 S \sin \omega t \cos \alpha \\ &= B_0 S \sin \omega t \cos \omega t = \frac{1}{2} B_0 S \sin 2\omega t. \end{aligned}$$

Hence,

$$\begin{aligned} \mathcal{V}_t' &= -\frac{d\Phi}{dt} = -\frac{d}{dt} \left( \frac{1}{2} B_0 S \sin 2\omega t \right), \\ &= -B_0 S \omega \cos 2\omega t \end{aligned}$$

as before.

## REVIEW QUESTIONS

- Q.6-1** What constitutes an *electromagnetostatic field*? In what ways are  $\mathbf{E}$  and  $\mathbf{B}$  related in a conducting medium under static conditions?
- Q.6-2** Write the fundamental postulate for electromagnetic induction.
- Q.6-3** State Lenz's law.
- Q.6-4** Write the expression for transformer emf.
- Q.6-5** How do the primary-to-secondary voltage and current ratios in an ideal transformer depend on the turns ratio of its windings?
- Q.6-6** What are *eddy currents*?
- Q.6-7** What is the principle of induction heating?
- Q.6-8** Why are materials having high permeability and low conductivity preferred as transformer cores?
- Q.6-9** Why are the cores of power transformers laminated?
- Q.6-10** Write the general form of Faraday's law.
- Q.6-11** What is a Faraday disk generator?

**REMARKS**

1.  $\mathbf{E}$  in a region of time-varying magnetic field is *not* conservative, and cannot be expressed as the gradient of a scalar potential alone.
2. A time-varying magnetic field linking with a circuit induces an emf in the circuit.
3. Transformers are inherently a-c devices.
4. An ideal transformer assumes infinite permeability for its core and infinite inductances for its windings.
5. The insulated laminations of a transformer core to reduce eddy-current power loss should be parallel to the direction of the magnetic flux.

**6-3 MAXWELL'S EQUATIONS**

The fundamental postulate for electromagnetic induction assures us that a time-varying magnetic field gives rise to an electric field. This assurance has been amply verified by numerous experiments. The  $\nabla \times \mathbf{E} = 0$  equation (6-1) must therefore be replaced by Eq. (6-7) in the time-varying case. We now have the following collection of two curl equations, (6-7) and (6-5), and two divergence equations, (6-2) and (6-4):

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}, \quad (6-7)(6-40a)$$

$$\nabla \times \mathbf{H} = \mathbf{J}, \quad (6-5)(6-40b)$$

$$\nabla \cdot \mathbf{D} = \rho_v, \quad (6-2)(6-40c)$$

$$\nabla \cdot \mathbf{B} = 0. \quad (6-4)(6-40d)$$

In addition, we know that the principle of conservation of charge must be satisfied at all times. The mathematical expression of charge conservation is the equation of continuity:

$$\nabla \cdot \mathbf{J} = -\frac{\partial \rho_v}{\partial t}. \quad (4-20)(6-41)$$

The crucial question here is whether the set of four equations in (6-40a, b, c, and d) is now consistent with the requirement specified by Eq. (6-41) in a time-varying situation. That the answer is in the negative is immediately obvious by simply taking the divergence of Eq. (6-40b),

$$\nabla \cdot (\nabla \times \mathbf{H}) = 0 = \nabla \cdot \mathbf{J}. \quad (6-42)$$

Equation (6-42) follows from the null identity, Eq. (2-109), which asserts that

the divergence of the curl of any well-behaved vector field is zero. Since Eq. (6-41) indicates that  $\nabla \cdot \mathbf{J}$  does not vanish in a time-varying situation, Eq. (6-40b) is, in general, not true.

How should Eqs. (6-40a, b, c, and d) be modified so that they are consistent with Eq. (6-41)? First of all, a term  $\partial \rho_v / \partial t$  must be added to the right side of Eq. (6-42):

$$\nabla \cdot (\nabla \times \mathbf{H}) = 0 = \nabla \cdot \mathbf{J} + \frac{\partial \rho_v}{\partial t}. \quad (6-43)$$

Using Eq. (6-40c) in Eq. (6-43), we have

$$\nabla \cdot (\nabla \times \mathbf{H}) = \nabla \cdot \left( \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t} \right),$$

which implies that

$$\nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t}. \quad (6-44)$$

Equation (6-44) indicates that a time-varying electric field will give rise to a magnetic field, even in the absence of a free current flow (i.e., even when  $\mathbf{J} = 0$ ). The additional term  $\partial \mathbf{D} / \partial t$  is necessary to make Eq. (6-44) consistent with the principle of conservation of charge.

It is easy to verify that  $\partial \mathbf{D} / \partial t$  has the dimension of a current density (SI unit: A/m<sup>2</sup>). The term  $\partial \mathbf{D} / \partial t$  is called **displacement current density**, and its introduction in the  $\nabla \times \mathbf{H}$  equation was one of the major contributions of James Clerk Maxwell (1831–1879). In order to be consistent with the equation of continuity in a time-varying situation, both of the curl equations in (6-1) and (6-5) must be generalized. The set of four consistent equations to replace the inconsistent equations in Eqs. (6-40a, b, c, and d) is

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}, \quad (6-45a)$$

$$\nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t}, \quad (6-45b)$$

$$\nabla \cdot \mathbf{D} = \rho_v, \quad (6-45c)$$

$$\nabla \cdot \mathbf{B} = 0. \quad (6-45d)$$

They are known as **Maxwell's equations**. Note that  $\rho_v$  in Eq. (6-45c) is the volume density of *free charges*, and  $\mathbf{J}$  in Eq. (6-45b) is the density of *free currents*, which may comprise both convection current ( $\rho_v \mathbf{u}$ ) and conduction current ( $\sigma \mathbf{E}$ ). These four equations, together with the equation of continuity in Eq. (6-41) and Lorentz's force equation in Eq. (5-5), form the foundation of

Time rate of change of  $\mathbf{D}$  gives rise to a displacement current density.

Differential (operator) form of Maxwell's equations

electromagnetic theory. These equations can be used to explain and predict *all* macroscopic electromagnetic phenomena.

The four Maxwell's equations are not all independent.

Although the four Maxwell's equations in Eqs. (6-45a, b, c, and d) are consistent, they are not all independent. As a matter of fact, the two divergence equations, Eqs. (6-45c and d), can be derived from the two curl equations, Eqs. (6-45a and b), by making use of the equation of continuity, Eq. (6-41). See Exercise 6.4 below.

#### ■ EXERCISE 6.4

Take the divergence of the two curl equations (6-45a) and (6-45b) and derive the two divergence equations (6-45c) and (6-45d) with the aid of the equation of continuity (6-41).

### 6-3.1 INTEGRAL FORM OF MAXWELL'S EQUATIONS

The four Maxwell's equations in (6-45a, b, c, and d) are differential equations that are valid at every point in space. In explaining electromagnetic phenomena in a physical environment we must deal with finite objects of specified shapes and boundaries. It is convenient to convert the differential forms into their integral-form equivalents. We take the surface integral of both sides of the curl equations in Eqs. (6-45a) and (6-45b) over an open surface  $S$  with a contour  $C$  and apply Stokes's theorem to obtain

Integral form of Maxwell's equations

$$\oint_C \mathbf{E} \cdot d\boldsymbol{\ell} = - \int_S \frac{\partial \mathbf{B}}{\partial t} \cdot d\mathbf{s} \quad (6-46a)$$

and

$$\oint_C \mathbf{H} \cdot d\boldsymbol{\ell} = \int_S \left( \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t} \right) \cdot d\mathbf{s}. \quad (6-46b)$$

Taking the volume integral of both sides of the divergence equations in Eqs. (6-45c) and (6-45d) over a volume  $V$  with a *closed* surface  $S$  and using divergence theorem, we have

$$\oint_S \mathbf{D} \cdot d\mathbf{s} = \int_V \rho_v dv \quad (6-46c)$$

and

$$\oint_S \mathbf{B} \cdot d\mathbf{s} = 0. \quad (6-46d)$$



TABLE 6-1 MAXWELL'S EQUATIONS AND THEIR SIGNIFICANCE

| Differential Form                                                                | Integral Form                                                                                                       | Significance                |
|----------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|-----------------------------|
| $\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$             | $\oint_C \mathbf{E} \cdot d\boldsymbol{\ell} = -\frac{d\Phi}{dt}$                                                   | Faraday's law               |
| $\nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t}$ | $\oint_C \mathbf{H} \cdot d\boldsymbol{\ell} = I + \int_S \frac{\partial \mathbf{D}}{\partial t} \cdot d\mathbf{s}$ | Ampère's circuital law      |
| $\nabla \cdot \mathbf{D} = \rho_v$                                               | $\oint_S \mathbf{D} \cdot d\mathbf{s} = Q$                                                                          | Gauss's law                 |
| $\nabla \cdot \mathbf{B} = 0$                                                    | $\oint_S \mathbf{B} \cdot d\mathbf{s} = 0$                                                                          | No isolated magnetic charge |

The set of four equations in (6-46a, b, c, and d) is the integral form of Maxwell's equations. Both the differential and integral forms of Maxwell's equations as well as their significance are listed in Table 6-1 for easy reference.

### EXAMPLE 6-6

An a-c source of amplitude  $V_0$  and angular frequency  $\omega$ ,  $v_c = V_0 \sin \omega t$ , is connected across a parallel-plate capacitor  $C_1$ , as shown in Fig. 6-7. (a) Verify that the displacement current in the capacitor is the same as the conduction current in the wires. (b) Determine the magnetic field intensity at a distance  $r$  from the wire.

### SOLUTION

a) The conduction current in the connecting wire is

$$i_C = C_1 \frac{dv_C}{dt} = C_1 V_0 \omega \cos \omega t.$$

For a parallel-plate capacitor with an area  $A$ , plate separation  $d$ , and a dielectric medium of permittivity  $\epsilon$ , the capacitance is

$$C_1 = \epsilon \frac{A}{d}.$$

With a voltage  $v_C$  appearing between the plates, the uniform electric field intensity  $E$  in the dielectric is equal to (neglecting fringing effects)  $E = v_C/d$ , whence

$$D = \epsilon E = \epsilon \frac{V_0}{d} \sin \omega t.$$

The displacement current is then

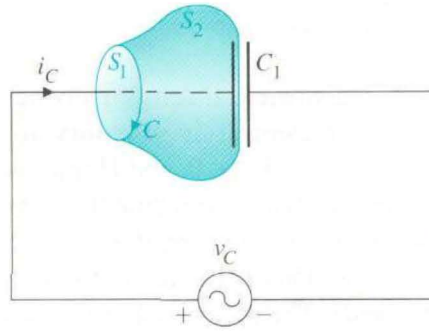


FIGURE 6-7 A parallel-plate capacitor connected to an a-c voltage source (Example 6-6).

$$\begin{aligned} i_D &= \int_A \frac{\partial \mathbf{D}}{\partial t} \cdot d\mathbf{s} = \left( \epsilon \frac{A}{d} \right) V_0 \omega \cos \omega t \\ &= C_1 V_0 \omega \cos \omega t = i_C, \end{aligned}$$

which is what we wanted to verify.

- b) The magnetic field intensity at a distance  $r$  from the conducting wire can be found by applying the generalized Ampère's circuital law, Eq. (6-46b), to contour  $C$  in Fig. 6-7. Two typical open surfaces with rim  $C$  may be chosen: (1) a planar disk surface  $S_1$ , or (2) a curved surface  $S_2$  passing through the dielectric medium. Symmetry around the wire ensures a constant  $H_\phi$  along the contour  $C$ . The line integral on the left side of Eq. (6-46b) is

$$\oint_C \mathbf{H} \cdot d\boldsymbol{\ell} = 2\pi r H_\phi.$$

For the surface  $S_1$ , only the first term on the right side of Eq. (6-46b) is nonzero because no charges are deposited along the wire and, consequently,  $\mathbf{D} = 0$ .

$$\int_{S_1} \mathbf{J} \cdot d\mathbf{s} = i_C = C_1 V_0 \omega \cos \omega t.$$

Since the surface  $S_2$  passes through the dielectric medium, no conduction current flows through  $S_2$ . If the second surface integral were not there, the right side of Eq. (6-46b) would be zero. This would result in a contradiction. The inclusion of the displacement-current term by Maxwell eliminates this contradiction. As we have shown in part (a),  $i_D = i_C$ . Hence we obtain the same result whether surface  $S_1$  or surface  $S_2$  is chosen. Equating the two previous integrals, we find that

$$H_\phi = \frac{C_1 V_0}{2\pi r} \omega \cos \omega t.$$

## 6-3.2 ELECTROMAGNETIC BOUNDARY CONDITIONS

In order to solve electromagnetic problems involving contiguous regions of different constitutive parameters, it is necessary to know the boundary conditions that the field vectors  $\mathbf{E}$ ,  $\mathbf{D}$ ,  $\mathbf{B}$ , and  $\mathbf{H}$  must satisfy at the interfaces. Boundary conditions are derived by applying the integral form of Maxwell's equations (6-46a, b, c, and d) to a small region at an interface of two media in manners similar to those used in obtaining the boundary conditions for static electric and magnetic fields. The integral equations are assumed to hold for regions containing discontinuous media. (You should review the procedures followed in Sections 3-8 and 5-9.) In general, the application of the integral form of a curl equation to a flat closed path at a boundary with top and bottom sides in the two touching media yields the boundary condition for the tangential components. On the other hand, the application of the integral form of a divergence equation to a shallow pillbox at an interface with top and bottom faces in the two contiguous media gives the boundary condition for the normal components.

The boundary conditions for the tangential components of  $\mathbf{E}$  and  $\mathbf{H}$  are obtained from Eqs. (6-46a) and (6-46b), respectively:

Boundary condition  
for tangential  
component of  $\mathbf{E}$

$$E_{1t} = E_{2t} \quad (\text{V/m}); \quad (6-47a)$$

Boundary condition  
for tangential  
component of  $\mathbf{H}$

$$\mathbf{a}_{n2} \times (\mathbf{H}_1 - \mathbf{H}_2) = \mathbf{J}_s \quad (\text{A/m}). \quad (6-47b)$$

We note that Eqs. (6-47a) and (6-47b) for the time-varying case are exactly the same as Eq. (3-72) for static electric fields and Eq. (5-71) for static magnetic fields, respectively, in spite of the existence of the time-varying terms in Eqs. (6-46a) and (6-46b).

Similarly, the boundary conditions for the normal components of  $\mathbf{D}$  and  $\mathbf{B}$  are obtained from Eqs. (6-46c) and (6-46d):

Boundary condition  
for normal  
component of  $\mathbf{D}$

$$D_{1n} - D_{2n} = \rho_s \quad (\text{C/m}^2), \quad (6-47c)$$

where the reference normal direction is outward from medium 2; and

Boundary condition  
for normal  
component of  $\mathbf{B}$

$$B_{1n} = B_{2n} \quad (\text{T}). \quad (6-47d)$$

These are the same as, respectively, Eq. (3-75b) for static electric fields and Eq.

(5-68) for static magnetic fields because we start from the same divergence equations.

In the following we summarize the boundary conditions of two important special cases.

**A. Interface between two lossless media**

A lossless linear medium can be specified by a permittivity  $\epsilon$  and a permeability  $\mu$ , with  $\sigma = 0$ . There are usually no free charges and no surface currents at the interface between two lossless media. We set  $\rho_s = 0$  and  $\mathbf{J}_s = 0$  in Eqs. (6-47c) and (6-47b), and obtain the boundary conditions listed in Table 6-2. We see that in this case  $E_t$ ,  $H_t$ ,  $D_n$  and  $B_n$  are all continuous at the interface.

**B. Interface between a dielectric and a perfect conductor**

A perfect conductor is one with an infinite conductivity. In the physical world we have an abundance of "good conductors" such as silver, copper, gold, and aluminium, whose conductivities are of the order of  $10^7$  (S/m). (See the table in Appendix B-4.) In order to simplify the analytical solution of field problems, good conductors are often considered perfect conductors in regard to boundary conditions. In the *interior* of a perfect conductor the electric field is zero (otherwise, it would produce an infinite current density), and any charges the conductor will have will reside on the surface only. The interrelationship between  $(\mathbf{E}, \mathbf{D})$  and  $(\mathbf{B}, \mathbf{H})$  through Maxwell's equations ensures that  $\mathbf{B}$  and  $\mathbf{H}$  are also zero in the *interior* of a conductor in a *time-varying situation*. Consider an interface between a lossless dielectric (medium 1) and a perfect conductor (medium 2). In medium 2,  $\mathbf{E}_2 = 0$ ,  $\mathbf{H}_2 = 0$ ,  $\mathbf{D}_2 = 0$ , and  $\mathbf{B}_2 = 0$ . The general boundary conditions in Eqs. (6-47a, b, c, and d) reduce to those listed in Table 6-3. In this case,  $E_t$  and  $B_n$  are continuous, but  $H_t$  and  $D_n$  are discontinuous by an amount equal to the surface current density  $J_s$  and surface charge density  $\rho_s$ , respectively. It is important to note that, when we apply Eqs. (6-49b) and (6-49c), the reference normal is outward from the surface of the conductor (medium 2).

TABLE 6-2 BOUNDARY CONDITIONS BETWEEN TWO LOSSLESS MEDIA

|                                                                                         |         |
|-----------------------------------------------------------------------------------------|---------|
| $E_{1t} = E_{2t} \longrightarrow \frac{D_{1t}}{D_{2t}} = \frac{\epsilon_1}{\epsilon_2}$ | (6-48a) |
| $H_{1t} = H_{2t} \longrightarrow \frac{B_{1t}}{B_{2t}} = \frac{\mu_1}{\mu_2}$           | (6-48b) |
| $D_{1n} = D_{2n} \longrightarrow \epsilon_1 E_{1n} = \epsilon_2 E_{2n}$                 | (6-48c) |
| $B_{1n} = B_{2n} \longrightarrow \mu_1 H_{1n} = \mu_2 H_{2n}$                           | (6-48d) |



TABLE 6-3 BOUNDARY CONDITIONS BETWEEN A DIELECTRIC (MEDIUM 1) AND A PERFECT CONDUCTOR (MEDIUM 2) (TIME-VARYING CASE)

| On the Side of Medium 1                              | On the Side of Medium 2 |         |
|------------------------------------------------------|-------------------------|---------|
| $E_{1t} = 0$                                         | $E_{2t} = 0$            | (6-49a) |
| $\mathbf{a}_{n2} \times \mathbf{H}_1 = \mathbf{J}_s$ | $H_{2t} = 0$            | (6-49b) |
| $D_{1n} = \rho_s$                                    | $D_{2n} = 0$            | (6-49c) |
| $B_{1n} = 0$                                         | $B_{2n} = 0$            | (6-49d) |

## ■ EXERCISE 6.5

Assume that the  $y = 0$  plane separates air in the upper half-space ( $y > 0$ ) from a good conductor and that a surface charge density  $\rho_s \approx C_1 \sin \beta x$  and a surface current density  $\mathbf{J}_s = \mathbf{a}_x C_2 \cos \beta x$  exist on the interface. ( $C_1$ ,  $C_2$ , and  $\beta$  are constants.) Determine  $\mathbf{E}$  and  $\mathbf{H}$  in the air at the interface.

ANS.  $\mathbf{E} = \mathbf{a}_y (C_1 / \epsilon_0) \sin \beta x$ ,  $\mathbf{H} = \mathbf{a}_z C_2 \cos \beta x$ .

## REVIEW QUESTIONS

**Q.6-12** Write the differential form of Maxwell's equations.

**Q.6-13** Explain the significance of *displacement current*.

**Q.6-14** Write the integral form of Maxwell's equations, and identify each equation with the proper experimental law.

**Q.6-15** State the boundary conditions for  $E_t$  and for  $B_n$ .

**Q.6-16** State the boundary conditions for  $H_t$  and for  $D_n$ .

**Q.6-17** Why is the  $\mathbf{E}$  field immediately outside of a perfect conductor perpendicular to the conductor surface?

**Q.6-18** Why is the  $\mathbf{H}$  field immediately outside of a perfect conductor tangential to the conductor surface?

**Q.6-19** Can a static magnetic field exist in the interior of a perfect conductor? Explain. Can a time-varying magnetic field? Explain.

## REMARKS

1. A changing magnetic field produces an electric field, and a changing electric field produces a displacement current that contributes to the magnetic field. In time-varying situations the electric and magnetic fields are coupled through Maxwell's curl equations.
2. Not all four Maxwell's equations are independent.
3. The tangential component of  $\mathbf{E}$  and the normal component of  $\mathbf{B}$  are continuous at the interface of any two media.

## 6-4 POTENTIAL FUNCTIONS

In Section 5-3 the concept of the vector magnetic potential  $\mathbf{A}$  was introduced because of the solenoidal nature of  $\mathbf{B}$  ( $\nabla \cdot \mathbf{B} = 0$ ):

$$\mathbf{B} = \nabla \times \mathbf{A} \quad (\text{T}). \quad (6-50)$$

If Eq. (6-50) is substituted in the differential form of Faraday's law, Eq. (6-7), we get

$$\nabla \times \mathbf{E} = -\frac{\partial}{\partial t} (\nabla \times \mathbf{A}), \quad (6-51)$$

or

$$\nabla \times \left( \mathbf{E} + \frac{\partial \mathbf{A}}{\partial t} \right) = 0. \quad (6-52)$$

Since the sum of the two vector quantities in the parentheses of Eq. (6-52) is curl-free, it can be expressed as the gradient of a scalar. To be consistent with the definition of the scalar electric potential  $V$  in Eq. (3-26) for electrostatics, we write

$$\mathbf{E} + \frac{\partial \mathbf{A}}{\partial t} = -\nabla V,$$

from which we obtain

$$\mathbf{E} = -\nabla V - \frac{\partial \mathbf{A}}{\partial t} \quad (\text{V/m}). \quad (6-53)$$

In time-varying case,  $\mathbf{E}$  is a function of both scalar electric potential  $V$  and vector magnetic potential  $\mathbf{A}$ .

In the static case,  $\partial \mathbf{A} / \partial t = 0$ , and Eq. (6-53) reduces to  $\mathbf{E} = -\nabla V$ . Hence  $\mathbf{E}$  can be determined from  $V$  alone, and  $\mathbf{B}$  from  $\mathbf{A}$  by Eq. (6-50). For time-varying fields,  $\mathbf{E}$  depends on both  $V$  and  $\mathbf{A}$ ; that is, an electric field intensity can result both from accumulations of charge through the  $-\nabla V$  term and from time-varying magnetic fields through the  $-\partial \mathbf{A} / \partial t$  term. Inasmuch as  $\mathbf{B}$  also depends on  $\mathbf{A}$ ,  $\mathbf{E}$  and  $\mathbf{B}$  are coupled.

Let us substitute Eqs. (6-50) and (6-53) into Eq. (6-45b) and make use of the constitutive relations  $\mathbf{H} = \mathbf{B} / \mu$  and  $\mathbf{D} = \epsilon \mathbf{E}$ . We have

$$\nabla \times \nabla \times \mathbf{A} = \mu \mathbf{J} + \mu \epsilon \frac{\partial}{\partial t} \left( -\nabla V - \frac{\partial \mathbf{A}}{\partial t} \right), \quad (6-54)$$

where a homogeneous medium has been assumed. Recalling the vector

identity for  $\nabla \times \nabla \times \mathbf{A}$  in Eq. (5-16a), we can write Eq. (6-54) as

$$\nabla(\nabla \cdot \mathbf{A}) - \nabla^2 \mathbf{A} = \mu \mathbf{J} - \nabla \left( \mu \epsilon \frac{\partial V}{\partial t} \right) - \mu \epsilon \frac{\partial^2 \mathbf{A}}{\partial t^2},$$

or

$$\nabla^2 \mathbf{A} - \mu \epsilon \frac{\partial^2 \mathbf{A}}{\partial t^2} = -\mu \mathbf{J} + \nabla \left( \nabla \cdot \mathbf{A} + \mu \epsilon \frac{\partial V}{\partial t} \right). \quad (6-55)$$

Now, the definition of a vector requires the specification of both its curl and its divergence. Although the curl of  $\mathbf{A}$  is designated  $\mathbf{B}$  in Eq. (6-50), we are still at liberty to choose the divergence of  $\mathbf{A}$ . We let

Lorentz condition for potentials

$$\nabla \cdot \mathbf{A} + \mu \epsilon \frac{\partial V}{\partial t} = 0, \quad (6-56)$$

which makes the second term on the right side of Eq. (6-55) vanish and reduces Eq. (6-55) to a simplest possible form. We then have

Wave equation for vector potential  $\mathbf{A}$

$$\nabla^2 \mathbf{A} - \mu \epsilon \frac{\partial^2 \mathbf{A}}{\partial t^2} = -\mu \mathbf{J}. \quad (6-57)$$

Equation (6-57) is the *nonhomogeneous wave equation for vector potential  $\mathbf{A}$* . It is called a wave equation because its solutions represent waves traveling with a velocity equal to  $1/\sqrt{\mu\epsilon}$ . This will become clear in Subsection 6-4.1 when the solution of wave equations is discussed. The relation between  $\mathbf{A}$  and  $V$  in Eq. (6-56) is called the *Lorentz condition (or Lorentz gauge) for potentials*. It reduces to the condition  $\nabla \cdot \mathbf{A} = 0$  in Eq. (5-19) for static fields.

A corresponding wave equation for the scalar potential  $V$  is

Wave potential for scalar potential  $V$

$$\nabla^2 V - \mu \epsilon \frac{\partial^2 V}{\partial t^2} = -\frac{\rho_v}{\epsilon}, \quad (6-58)$$

which is the *nonhomogeneous wave equation for scalar potential  $V$* . Hence the Lorentz condition in Eq. (6-56) uncouples the wave equations for  $\mathbf{A}$  and for  $V$ . Note the similarities of Eqs. (6-57) and (6-58), and the analogy among the quantities:  $\mathbf{A} \sim V$ ,  $\mathbf{J} \sim \rho_v$ , and  $\mu \sim 1/\epsilon$ .

### ■ EXERCISE 6.6

Derive the wave equation (6-58) for  $V$  by using Eq. (6-53) in Eq. (6-40c).

## REVIEW QUESTIONS

**Q.6-20** How are the scalar electric potential  $V$  and the vector magnetic potential  $\mathbf{A}$  defined?

**Q.6-21** What is a wave equation?

## REMARKS

Time-varying potential functions  $V$  and  $\mathbf{A}$  satisfy the wave equation.

## 6-4.1 SOLUTION OF WAVE EQUATIONS

We now consider the solution of the nonhomogeneous wave equation (6-58) for scalar potential  $V$  due to a charge distribution  $\rho_v$  in a restricted region. Let an elemental point charge  $\rho_v dv'$  be situated at the origin at time  $t$ . At a distance  $R$  far away from the origin we can assume spherical symmetry (that is,  $V$  depends only on  $R$  and  $t$ , not on  $\theta$  or  $\phi$ ). Using Eq. (3-129), we can write Eq. (6-58) as

$$\frac{1}{R^2} \frac{\partial}{\partial R} \left( R^2 \frac{\partial V}{\partial R} \right) - \mu\epsilon \frac{\partial^2 V}{\partial t^2} = 0. \quad (6-59)$$

We now introduce a new variable

$$V(R, t) = \frac{1}{R} U(R, t), \quad (6-60)$$

which simplifies Eq. (6-59) to

$$\frac{\partial^2 U}{\partial R^2} - \mu\epsilon \frac{\partial^2 U}{\partial t^2} = 0. \quad (6-61)$$

Equation (6-61) is a one-dimensional homogeneous wave equation. It can be verified by direct substitution (see Problem P.6-11) that *any* twice-differentiable function of  $(t - R\sqrt{\mu\epsilon})$  is a solution of Eq. (6-61). We write

$$U(R, t) = f(t - R\sqrt{\mu\epsilon}). \quad (6-62)$$

The function at new distance  $R + \Delta R$  at a later time  $t + \Delta t$  is

$$U(R + \Delta R, t + \Delta t) = f[t + \Delta t - (R + \Delta R)\sqrt{\mu\epsilon}]$$

which equals  $f(t - R\sqrt{\mu\epsilon})$  and retains its form if  $\Delta t = \Delta R\sqrt{\mu\epsilon} = \Delta R/u_p$ . The



Velocity of wave propagation in a medium with constitutive parameters  $\epsilon$  and  $\mu$  is  $1/\sqrt{\mu\epsilon}$ .

quantity

$$u_p = \frac{1}{\sqrt{\mu\epsilon}} \quad (6-63)$$

is the **velocity of wave propagation**, a characteristic of the medium. In air, it is equal to  $c = 1/\sqrt{\mu_0\epsilon_0}$ . From Eqs. (6-60) and (6-62) we have

$$V(R, t) = \frac{1}{R} f(t - R/u_p). \quad (6-64)$$

To determine what the specific function  $f(t - R/u_p)$  must be, we note from Eq. (3-29) that for a static point charge  $\rho(t) \Delta v'$  at the origin,

$$\Delta V(R) = \frac{\rho_v(t) \Delta v'}{4\pi\epsilon R}. \quad (6-65)$$

Comparison of Eqs. (6-64) and (6-65) enables us to identify

$$\Delta f(t - R/u_p) = \frac{\rho_v(t - R/u_p) \Delta v'}{4\pi\epsilon}. \quad (6-66)$$

The potential due to a charge distribution over a volume  $V'$  is then

$$V(R, t) = \frac{1}{4\pi\epsilon} \int_{V'} \frac{\rho_v(t - R/u_p)}{R} dv' \quad (\text{V}). \quad (6-67)$$

Equation (6-67) indicates that the scalar potential at a distance  $R$  from the source at time  $t$  depends on the value of the charge density at an *earlier* time  $(t - R/u_p)$ . For this reason,  $V(R, t)$  in Eq. (6-67) is called the **retarded scalar potential**.

The solution of the nonhomogeneous wave equation, Eq. (6-57), for vector magnetic potential  $\mathbf{A}$  can proceed in exactly the same way as that for  $V$ . The vector equation, Eq. (6-57), for  $\mathbf{A}$  can be decomposed into three scalar equations, each similar to Eq. (6-58) for  $V$ . The **retarded vector potential** is thus given by

$$\mathbf{A}(R, t) = \frac{\mu}{4\pi} \int_{V'} \frac{\mathbf{J}(t - R/u_p)}{R} dv' \quad (\text{Wb/m}). \quad (6-68)$$

The electric and magnetic fields derived from  $\mathbf{A}$  and  $V$  by differentiation will obviously also be functions of  $(t - R/u_p)$  and therefore retarded in time. It takes time for electromagnetic waves to travel and for the effects of time-varying charges and currents to be felt at distant points. In circuit theory this time-retardation effect is ignored and instant response is assumed.

Finding retarded scalar potential  $V$  from charge distribution

Finding retarded vector potential  $\mathbf{A}$  from current distribution

Circuit theory ignores time-retardation effect

## ■ EXERCISE 6.7

A radar signal sent from earth to the moon is received back on the earth after a delay of 2.562 (s). Determine the distance between the surfaces of earth and the moon at that moment in kilometers and in miles.

ANS.  $3.843 \times 10^5$  (km), or 238,844 miles.

## ■ EXERCISE 6.8

What is the distance equivalent to one light year?

ANS.  $9.46 \times 10^{12}$  (km), or 5.88 trillion miles.

## REVIEW QUESTIONS

**Q.6-22** What does *retarded potential* mean in electromagnetics?

**Q.6-23** In what ways do the retardation time and the velocity of wave propagation depend on the constitutive parameters of the medium?

## REMARKS

1. The response at a distance of changing charge and current distributions is *not instantaneous*, but is *delayed* in time.
2. The velocity of wave propagation is a characteristic of the medium and is *independent of frequency*.

## 6-5 TIME-HARMONIC FIELDS

Maxwell's equations and all the equations derived from them so far in this chapter hold for electromagnetic quantities with an arbitrary time-dependence. The actual type of time functions that the field quantities assume depends on the source functions  $\rho_v$  and  $\mathbf{J}$ . In engineering, sinusoidal time functions occupy a unique position. They are easy to generate; arbitrary periodic time functions can be expanded into Fourier series of harmonic sinusoidal components; and transient nonperiodic functions can be expressed as Fourier integrals. Hence for source functions with an arbitrary time dependence, electrodynamic fields can be determined in terms of those caused by the various frequency components of the source functions. The application of the principle of superposition (adding the results due to the various frequencies) will give us the total fields. In this section we examine *time-harmonic* (steady-state sinusoidal) field relationships.

Time-harmonic fields are sinusoidal fields.

## 6-5.1 THE USE OF PHASORS—A REVIEW

For time-harmonic fields it is convenient to use a phasor notation. At this time we digress briefly to review the use of phasors. Conceptually, it is simpler to discuss a scalar phasor. The instantaneous (time-dependent) expression of

a sinusoidal scalar quantity, such as a current  $i$ , can be written as either a cosine or a sine function. If we choose a cosine function as the *reference* (which is usually dictated by the functional form of the excitation), then all derived results will refer to the cosine function. The specification of a sinusoidal quantity requires the knowledge of three parameters: amplitude, frequency, and phase. For example,

$$i(t) = I_0 \cos(\omega t + \phi), \quad (6-69)$$

where  $I_0$  is the amplitude;  $\omega$  is the angular frequency (rad/s)— $\omega$  is always equal to  $2\pi f$ ,  $f$  being the frequency in hertz; and  $\phi$  is the phase angle referred to the cosine function. We can also write  $i(t)$  in Eq. (6-69) as a sine function if we wish:  $i(t) = I_0 \sin(\omega t + \phi')$ , with  $\phi' = \phi + \pi/2$ . Thus it is important to decide at the outset whether our reference is a cosine or a sine function, then to stick to that decision throughout a problem.

To work directly with an instantaneous expression such as the cosine function is inconvenient when differentiations or integrations of  $i(t)$  are involved because they lead to both sine (first-order differentiation or integration) and cosine (second-order differentiation or integration) functions and because it is tedious to combine sine and cosine functions. For instance, the loop equation for a series RLC circuit with an applied voltage  $v(t) = V_0 \cos \omega t$  is

$$L \frac{di}{dt} + Ri + \frac{1}{C} \int i dt = v(t). \quad (6-70)$$

If we write the resulting current  $i(t)$  in the form of Eq. (6-69), Eq. (6-70) yields

$$\begin{aligned} I_0 \left[ -\omega L \sin(\omega t + \phi) + R \cos(\omega t + \phi) + \frac{1}{\omega C} \sin(\omega t + \phi) \right] \\ = V_0 \cos \omega t. \end{aligned} \quad (6-71)$$

Obviously, complicated mathematical manipulations are required in order to determine the unknown  $I_0$  and  $\phi$  from Eq. (6-71).

It is much simpler to use exponential functions<sup>†</sup> by writing the applied voltage as

$$\begin{aligned} v(t) &= V_0 \cos \omega t = \Re e[(V_0 e^{j0})e^{j\omega t}] \\ &= \Re e[V_s e^{j\omega t}] \end{aligned} \quad (6-72)$$

and  $i(t)$  in Eq. (6-69) as

$$\begin{aligned} i(t) &= \Re e[(I e^{j\phi})e^{j\omega t}] \\ &= \Re e[I_s e^{j\omega t}], \end{aligned} \quad (6-73)$$

<sup>†</sup> $e^{j\omega t} = \cos \omega t + j \sin \omega t$ ,  $\cos \omega t = \Re e(e^{j\omega t})$ ,  $\sin \omega t = \Im m(e^{j\omega t})$ .



where  $\Re$  means “the real part of.” In Eqs. (6-72) and (6-73),

$$V_s = V_0 e^{j0} = V_0 \quad (6-74)$$

$$I_s = I_0 e^{j\phi} \quad (6-75)$$

are (scalar) **phasors** that contain amplitude and phase information but are *independent of  $t$* . The phasor  $V_s$  in Eq. (6-74) with zero phase angle is the reference phasor. From Eq. (6-73) we have

$$\frac{di}{dt} = \Re(j\omega I_s e^{j\omega t}), \quad \text{and} \quad (6-76)$$

$$\int i dt = \Re\left(\frac{I_s}{j\omega} e^{j\omega t}\right). \quad (6-77)$$

Substitution of Eqs. (6-72) through (6-77) in Eq. (6-70) yields

$$\left[R + j\left(\omega L - \frac{1}{\omega C}\right)\right] I_s = V_s, \quad (6-78)$$

from which the current phasor  $I_s$  can be solved easily. Note that the time-dependent factor  $e^{j\omega t}$  disappears from Eq. (6-78) because it is present in every term after the substitution in Eq. (6-70) and is therefore canceled. After  $I_s$  has been determined, the instantaneous current response  $i(t)$  can be found from Eq. (6-73) by multiplying  $I_s$  by  $e^{j\omega t}$  and taking the real part of the product.

If the applied voltage had been given as a *sine function* such as  $v(t) = V_0 \sin \omega t$ , the series RLC-circuit problem would be solved in terms of phasors in exactly the same way; only the instantaneous expressions would be obtained by taking the *imaginary part* of the product of the phasors with  $e^{j\omega t}$ . The complex phasors represent the magnitudes and the phase shifts of the quantities in the solution of time-harmonic problems.

Phasors: polar forms of complex quantities containing amplitude and phase information

Converting phasors to instantaneous sinusoidal time functions

### EXAMPLE 6-7

Write the phasor expression  $I_s$  for the following current functions using a cosine reference.

- a)  $i(t) = -I_0 \cos(\omega t - 30^\circ)$ , and
- b)  $i(t) = I_0 \sin(\omega t + 0.2\pi)$ .

#### SOLUTION

For cosine reference we write

$$i(t) = \Re(I_s e^{j\omega t}).$$

- a)  $i(t) = -I_0 \cos(\omega t - 30^\circ)$   
 $= \Re[(-I_0 e^{-j30^\circ}) e^{j\omega t}].$

$$\text{Thus, } I_s = -I_0 e^{-j30^\circ} = -I_0 e^{-j\pi/6} = I_0 e^{j5\pi/6}.$$



$$\begin{aligned}\text{b)} \quad i(t) &= I_0 \sin(\omega t + 0.2\pi) \\ &= \Re e[(I_0 e^{j0.2\pi})e^{-j\pi/2} \cdot e^{j\omega t}],\end{aligned}$$

where the  $e^{-j\pi/2}$  factor is needed because the phase of  $\sin \omega t$  lags behind that of  $\cos \omega t$  by  $90^\circ$  or  $\pi/2$  (rad). We have

$$I_s = (I_0 e^{j0.2\pi})e^{-j\pi/2} = I_0 e^{-j0.3\pi}.$$

### EXERCISE 6.9

Find the phasor expressions  $I'_s$  for the current functions in Example 6-7 using a sine reference,  $i(t) = \Im m(I'_s e^{j\omega t})$ .

$$\text{ANS. (a) } -I_0 e^{j\pi/3}, \text{ or } I_0 e^{-j2\pi/3}. \quad \text{(b) } I_0 e^{j0.2\pi}.$$

### EXAMPLE 6-8

Write the instantaneous expressions  $v(t)$  for the following phasors using a cosine reference:

$$\text{a)} \quad V_s = V_0 e^{j\pi/4}, \quad \text{and}$$

$$\text{b)} \quad V_s = 3 - j4.$$

#### SOLUTION

$$\begin{aligned}\text{a)} \quad v(t) &= \Re e[V_s e^{j\omega t}] \\ &= \Re e[(V_0 e^{j\pi/4})e^{j\omega t}] \\ &= V_0 \cos(\omega t + \pi/4).\end{aligned}$$

$$\begin{aligned}\text{b)} \quad V_s &= 3 - j4 = \sqrt{3^2 + 4^2} e^{j \tan^{-1}(-4/3)} \\ &= 5e^{-j53.1^\circ}.\end{aligned}$$

Thus,

$$\begin{aligned}v(t) &= \Re e[(5e^{-j53.1^\circ})e^{j\omega t}] \\ &= 5 \cos(\omega t - 53.1^\circ).\end{aligned}$$

### EXERCISE 6.10

Write the phasor expression  $V_s$  for the voltage  $v(t) = 10 \cos(\omega t - 45^\circ)$ .

$$\text{ANS. } 10e^{-j\pi/4}.$$

### EXERCISE 6.11

Write the instantaneous expression  $v(t)$  for the phasor  $V_s = 4 + j3$  using a cosine reference.

$$\text{ANS. } 5 \cos(\omega t + 36.9^\circ).$$

### EXERCISE 6.12

Write the instantaneous expression  $v(t)$  for the phasor  $V_s = 4 + j3$  using a sine reference.

$$\text{ANS. } 5 \sin(\omega t + 126.9^\circ).$$

## REVIEW QUESTIONS

**Q.6-24** What is a *phasor*? Are phasors functions of  $t$ ? Functions of  $\omega$ ?

**Q.6-25** What is the difference between a phasor and a vector?

## REMARKS

1. Phasors are complex quantities (expressed in polar form) representing the magnitude and the phase of sinusoidal functions.
2. Phase angles can be expressed either in radians or in degrees. *Do not forget the  $^\circ$  sign when expressed in degrees.*
3. *Never mix factors containing  $j$  with instantaneous time functions.* Expressions such as  $j \cos \omega t$ ,  $e^{j\phi} \sin \omega t$ , and  $(1 - j)i(t)$  are incorrect.

## 6-5.2 TIME-HARMONIC ELECTROMAGNETICS

Field vectors that vary with space coordinates and are sinusoidal functions of time can similarly be represented by vector phasors that depend on space coordinates but not on time. As an example, we can write a time-harmonic  $\mathbf{E}$  field referring to  $\cos \omega t$ <sup>\*</sup> as

$$\mathbf{E}(x, y, z; t) = \Re \{ \mathbf{E}(x, y, z) e^{j\omega t} \}, \quad (6-79)$$

where  $\mathbf{E}(x, y, z)$  is a **vector phasor** that contains information on direction, magnitude, and phase. From Eqs. (6-76) and (6-77) we see that, if  $\mathbf{E}(x, y, z; t)$  is to be represented by the vector phasor  $\mathbf{E}(x, y, z)$ , then  $\partial \mathbf{E}(x, y, z; t) / \partial t$  and  $\int \mathbf{E}(x, y, z; t) dt$  would be represented by vector phasors  $j\omega \mathbf{E}(x, y, z)$  and  $\mathbf{E}(x, y, z) / j\omega$ , respectively. Higher-order differentiations and integrations with respect to  $t$  would be represented by multiplications and divisions, respectively, of the phasor  $\mathbf{E}(x, y, z)$  by higher powers of  $j\omega$ .

We now write time-harmonic Maxwell's equations (6-45a, b, c, and d) in terms of vector field phasors ( $\mathbf{E}$ ,  $\mathbf{H}$ ) and source phasors ( $\rho_v$ ,  $\mathbf{J}$ ) in a simple (linear, isotropic, and homogeneous) medium as follows.

Time-harmonic  
Maxwell's equations  
in terms of phasors

$$\nabla \times \mathbf{E} = -j\omega\mu\mathbf{H}, \quad (6-80a)$$

$$\nabla \times \mathbf{H} = \mathbf{J} + j\omega\epsilon\mathbf{E}, \quad (6-80b)$$

$$\nabla \cdot \mathbf{E} = \rho_v / \epsilon, \quad (6-80c)$$

$$\nabla \cdot \mathbf{H} = 0. \quad (6-80d)$$

The space-coordinate arguments and the subscript  $s$  indicating a phasor quantity have been omitted for simplicity. The fact that the same notations

<sup>\*</sup>If the time reference is not explicitly specified, it is customarily taken as  $\cos \omega t$ .

are used for the phasors as are used for their corresponding time-dependent quantities should create little confusion because we will deal almost exclusively with time-harmonic fields (and therefore with phasors) in the rest of this book. When there is a need to distinguish an instantaneous quantity from a phasor, the time dependence of the instantaneous quantity will be indicated explicitly by the inclusion of a  $t$  in its argument.

### REVIEW QUESTION

**Q.6-26** Discuss the advantages of using phasors in electromagnetics.

### REMARKS

1. Phasor quantities are not functions of  $t$ .
2. Instantaneous time functions cannot contain complex numbers.
3. Any electromagnetic expression containing  $j$  must necessarily be a relation of phasors.

In terms of phasors we can write time-harmonic wave equation (6-58) for scalar potential  $V$  as

$$\nabla^2 V - \mu\epsilon(j\omega)^2 V = -\frac{\rho_v}{\epsilon},$$

or

Helmholtz's equation  
for scalar potential  $V$

$$\nabla^2 V + k^2 V = -\frac{\rho_v}{\epsilon}, \quad (6-81)$$

where

$$k = \omega\sqrt{\mu\epsilon} = \frac{\omega}{u_p}, \quad (6-82a)$$

or

$$k = \frac{2\pi f}{u_p} = \frac{2\pi}{\lambda} \quad (6-82b)$$

Definition of wave  
number

is called the **wavenumber**. It is a measure of the number of wavelengths in a cycle. Similarly, the phasor form of time-harmonic wave equation (6-57) for vector potential  $\mathbf{A}$  is

Helmholtz's equation  
for vector potential  $\mathbf{A}$

$$\nabla^2 \mathbf{A} + k^2 \mathbf{A} = -\mu \mathbf{J}. \quad (6-83)$$

Equations (6-81) and (6-83) are referred to as *nonhomogeneous Helmholtz's equations*.

■ **EXERCISE 6.13** Write the phasor form for the Lorentz condition for potentials, Eq. (6-56).

### REVIEW QUESTION

**Q.6-27** Define *wavenumber*. What is its SI unit?

### REMARKS

1. Helmholtz's equations are time-harmonic wave equations in terms of phasors.
2. Wavenumber depends on medium characteristics as well as on the frequency of the wave, but is always equal to  $2\pi$  divided by the wavelength.

The solution of the nonhomogeneous Helmholtz's equation (6-81) for  $V$  can be obtained from Eq. (6-67). The potential  $V(R, t)$  involves a time advance  $R/u_p$  for  $\rho_v$ , which is equivalent to a phase lead of  $\omega(R/u_p)$  or  $kR$ . In phasor notation this requires a multiplying factor  $e^{-jkR}$ . Hence the *phasor form* for Eq. (6-67) is

Phasor form for  
retarded scalar  
potential

$$V(R) = \frac{1}{4\pi\epsilon} \int_{V'} \frac{\rho_v e^{-jkR}}{R} dv' \quad (\text{V}). \quad (6-84)$$

Similarly, the phasor solution of Eq. (6-83) for  $\mathbf{A}$  is

Phasor form for  
retarded vector  
potential

$$\mathbf{A}(R) = \frac{\mu}{4\pi} \int_{V'} \frac{\mathbf{J} e^{-jkR}}{R} dv' \quad (\text{Wb/m}). \quad (6-85)$$

These are the expressions for the retarded scalar and vector potentials due to time-harmonic sources. Now the Taylor-series expansion for the exponential factor  $e^{-jkR}$  is

$$e^{-jkR} = 1 - jkR + \frac{k^2 R^2}{2} + \dots \quad (6-86)$$

Thus, if

$$kR = 2\pi \frac{R}{\lambda} \ll 1, \quad (6-87)$$

or if the distance  $R$  is very small in comparison to the wavelength  $\lambda$ ,  $e^{-jkR}$  can



Procedure for  
determining  
instantaneous  
electric and  
magnetic fields

be approximated by 1. Equations (6-84) and (6-85) then simplify to the static expressions in Eqs. (3-39) and (5-22), respectively.

The formal procedure for determining the electric and magnetic fields due to time-harmonic charge and current distributions is as follows:

1. Find phasors  $V(R)$  and  $\mathbf{A}(R)$  from Eqs. (6-84) and (6-85).
2. Find phasors  $\mathbf{E}(R) = -\nabla V - j\omega\mathbf{A}$  and  $\mathbf{B}(R) = \nabla \times \mathbf{A}$ .
3. Find instantaneous  $\mathbf{E}(R, t) = \Re e[\mathbf{E}(R)e^{j\omega t}]$  and  $\mathbf{B}(R, t) = \Re e[\mathbf{B}(R)e^{j\omega t}]$  for a cosine reference.

The degree of difficulty of a problem depends on how difficult it is to perform the integrations in Step 1. We will use this procedure to determine the radiation properties of antennas in Chapter 10.

### EXAMPLE 6-9

Given that the electric field intensity of an electromagnetic wave in a nonconducting dielectric medium with permittivity  $\epsilon = 9\epsilon_0$  and permeability  $\mu_0$  is

$$\mathbf{E}(z, t) = \mathbf{a}_y 5 \cos(10^9 t - \beta z) \quad (\text{V/m}), \quad (6-88)$$

find the magnetic field intensity  $\mathbf{H}$  and the value of  $\beta$ .

#### SOLUTION

The given  $\mathbf{E}(z, t)$  in Eq. (6-88) is a harmonic time function with angular frequency  $\omega = 10^9$  (rad/s). Using phasors with a cosine reference, we have

$$\mathbf{E}(z) = \mathbf{a}_y 5e^{-j\beta z}. \quad (6-89)$$

The magnetic field intensity can be calculated from Eq. (6-80a).

$$\begin{aligned} \mathbf{H}(z) &= -\frac{1}{j\omega\mu_0} \nabla \times \mathbf{E} \\ &= -\frac{1}{j\omega\mu_0} \begin{vmatrix} \mathbf{a}_x & \mathbf{a}_y & \mathbf{a}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 0 & 5e^{-j\beta z} & 0 \end{vmatrix} = -\frac{1}{j\omega\mu_0} \left( -\mathbf{a}_x \frac{\partial}{\partial z} E_y \right) \\ &= -\frac{1}{j\omega\mu_0} (\mathbf{a}_x j\beta 5e^{-j\beta z}) = -\mathbf{a}_x \frac{\beta}{\omega\mu_0} 5e^{-j\beta z}. \end{aligned} \quad (6-90)$$

In order to determine  $\beta$ , we use Eq. (6-80b). For a nonconducting medium,  $\sigma = 0$ ,  $\mathbf{J} = 0$ . Thus,

$$\begin{aligned} \mathbf{E}(z) &= \frac{1}{j\omega\epsilon} \nabla \times \mathbf{H} = \frac{1}{j\omega\epsilon} \left( \mathbf{a}_y \frac{\partial}{\partial z} H_x \right) \\ &= \mathbf{a}_y \frac{\beta^2}{\omega^2 \mu_0 \epsilon} 5e^{-j\beta z}. \end{aligned} \quad (6-91)$$

Equating Eqs. (6-89) and (6-91), we require

$$\begin{aligned}\beta &= \omega\sqrt{\mu_0\epsilon} = 3\omega\sqrt{\mu_0\epsilon_0} = \frac{3\omega}{c} \\ &= \frac{3 \times 10^9}{3 \times 10^8} = 10 \quad (\text{rad/m}).\end{aligned}$$

From Eq. (6-90) we obtain

$$\begin{aligned}\mathbf{H}(z) &= -\mathbf{a}_x \frac{5(10)}{(10^9)(4\pi 10^{-7})} e^{-j10z} \\ &= -\mathbf{a}_x 0.0398 e^{-j10z}.\end{aligned}\tag{6-92}$$

The phasor  $\mathbf{H}(z)$  in Eq. (6-92) corresponds to the following instantaneous time function:

$$\mathbf{H}(z, t) = -\mathbf{a}_x 0.0398 \cos(10^9 t - 10z) \quad (\text{A/m}).\tag{6-93}$$

### REVIEW QUESTIONS

**Q.6-28** Write the phasor expression of scalar electric potential  $V(R)$  in terms of charge distribution  $\rho_v$ .

**Q.6-29** Write the phasor expression of vector magnetic potential  $\mathbf{A}(R)$  in terms of current distribution  $\mathbf{J}$ .

### REMARKS

The electric and magnetic field intensities of an electromagnetic wave in a given medium are definitely related. They must satisfy Maxwell's equations, and their amplitudes and phases cannot be specified independently.

### 6-5.3 THE ELECTROMAGNETIC SPECTRUM

In *source-free nonconducting media* characterized by  $\epsilon$  and  $\mu(\sigma = 0)$  Maxwell's equations (6-45a, b, c, and d) reduce to

Maxwell's equations  
in source-free  
nonconducting  
media

$$\nabla \times \mathbf{E} = -\mu \frac{\partial \mathbf{H}}{\partial t},\tag{6-94a}$$

$$\nabla \times \mathbf{H} = \epsilon \frac{\partial \mathbf{E}}{\partial t},\tag{6-94b}$$

$$\nabla \cdot \mathbf{E} = 0,\tag{6-94c}$$

$$\nabla \cdot \mathbf{H} = 0.\tag{6-94d}$$

Equations (6-94a, b, c, and d) are first-order differential equations in the two variables  $\mathbf{E}$  and  $\mathbf{H}$ . They can be combined to give a second-order equation in  $\mathbf{E}$  or  $\mathbf{H}$  alone. To do this, we take the curl of Eq. (6-94a) and use Eq. (6-94b):

$$\nabla \times \nabla \times \mathbf{E} = -\mu \frac{\partial}{\partial t} (\nabla \times \mathbf{H}) = -\mu\epsilon \frac{\partial^2 \mathbf{E}}{\partial t^2}.$$

Now  $\nabla \times \nabla \times \mathbf{E} = \nabla(\nabla \cdot \mathbf{E}) - \nabla^2 \mathbf{E} = -\nabla^2 \mathbf{E}$  because of Eq. (6-94c). Hence we have

$$\nabla^2 \mathbf{E} - \mu\epsilon \frac{\partial^2 \mathbf{E}}{\partial t^2} = 0; \quad (6-95)$$

or, since  $u_p = 1/\sqrt{\mu\epsilon}$ ,

Homogeneous wave  
equation for  $\mathbf{E}$

$$\nabla^2 \mathbf{E} - \frac{1}{u_p^2} \frac{\partial^2 \mathbf{E}}{\partial t^2} = 0. \quad (6-96)$$

In an entirely similar way we can also obtain an equation in  $\mathbf{H}$ :

Homogeneous wave  
equation for  $\mathbf{H}$

$$\nabla^2 \mathbf{H} - \frac{1}{u_p^2} \frac{\partial^2 \mathbf{H}}{\partial t^2} = 0. \quad (6-97)$$

Equations (6-96) and (6-97) are *homogeneous vector wave equations*.

We can see that in Cartesian coordinates Eqs. (6-96) and (6-97) can each be decomposed into three one-dimensional, homogeneous, scalar wave equations. Each component of  $\mathbf{E}$  and of  $\mathbf{H}$  will satisfy an equation exactly like Eq. (6-61), whose solutions represent waves. We will extensively discuss wave behavior in various environments in the next chapter.

For time-harmonic fields it is convenient to use phasors. Thus Eq. (6-96) becomes

$$\nabla^2 \mathbf{E}_s + \frac{\omega^2}{u_p^2} \mathbf{E}_s = 0,$$

or

Homogeneous  
Helmholtz's equation  
for phasor  $\mathbf{E}_s$

$$\nabla^2 \mathbf{E}_s + k^2 \mathbf{E}_s = 0, \quad (6-98)^\dagger$$

in view of Eq. (6-82a). Similarly, Eq. (6-97) leads to

Homogeneous  
Helmholtz's equation  
for phasor  $\mathbf{H}_s$

$$\nabla^2 \mathbf{H}_s + k^2 \mathbf{H}_s = 0. \quad (6-99)^\dagger$$

<sup>†</sup>Subscript  $s$  has been added to emphasize that  $\mathbf{E}_s$  and  $\mathbf{H}_s$  are phasors and are not the same as the time-dependent  $\mathbf{E}$  and  $\mathbf{H}$  in Eqs. (6-96) and (6-97).



The solutions of Eqs. (6-98) and (6-99) represent propagating waves, which will be the subject of study of the next chapter.

### REVIEW QUESTION

**Q.6-30** Explain why there may be nonvanishing solutions for electric and magnetic fields in source-free regions.

### REMARKS

1. In source-free nonconducting media  $\mathbf{E}$  and  $\mathbf{H}$  satisfy the same homogeneous wave equation, in Eqs. (6-96) and (6-97).
2. For time-harmonic waves in source-free media phasors  $\mathbf{E}_s$  and  $\mathbf{H}_s$  satisfy the same homogeneous Helmholtz's equation, in Eqs. (6-98) and (6-99).
3. Maxwell's equations, and therefore the wave and Helmholtz's equations, impose no limit on the frequency of the waves.

The electromagnetic spectrum and applications

The visible-light spectrum

Wireless transmission

Microwave radar bands

The electromagnetic spectrum that has been investigated experimentally extends from very low power frequencies through radio, television, microwave, infrared, visible light, ultraviolet, X-ray, and gamma ( $\gamma$ )-ray frequencies exceeding  $10^{24}$  (Hz). Figure 6-8 shows the electromagnetic spectrum divided into frequency and wavelength ranges on logarithmic scales according to application and natural occurrence.

The term “microwave” is somewhat nebulous and imprecise; it could mean electromagnetic waves above a frequency of 1 (GHz) and all the way up to the lower limit of the infrared band, encompassing UHF, SHF, EHF, and mm-wave regions. The wavelength range of visible light is from deep red at 720 (nm) to violet at 380 (nm), or from  $0.72 (\mu\text{m})$  to  $0.38 (\mu\text{m})$ , corresponding to a frequency range of from  $4.2 \times 10^{14}$  (Hz) to  $7.9 \times 10^{14}$  (Hz). The bands used for radar, satellite communication, navigation aids, television (TV), FM and AM radio, citizen's band radio (CB), sonar, and others are also noted. Frequencies below the VLF range are seldom used for wireless transmission because huge antennas would be needed for efficient radiation of electromagnetic waves and because of the very low data rate at these low frequencies. There have been proposals to use these frequencies for strategic global communication with submarines submerged in conducting seawater. In radar work it has been found convenient to assign alphabet names to the different microwave frequency bands. They are listed in Table 6-4.

In the next chapter we shall discuss the characteristics of plane electromagnetic waves and examine their behavior as they propagate across discontinuous boundaries.



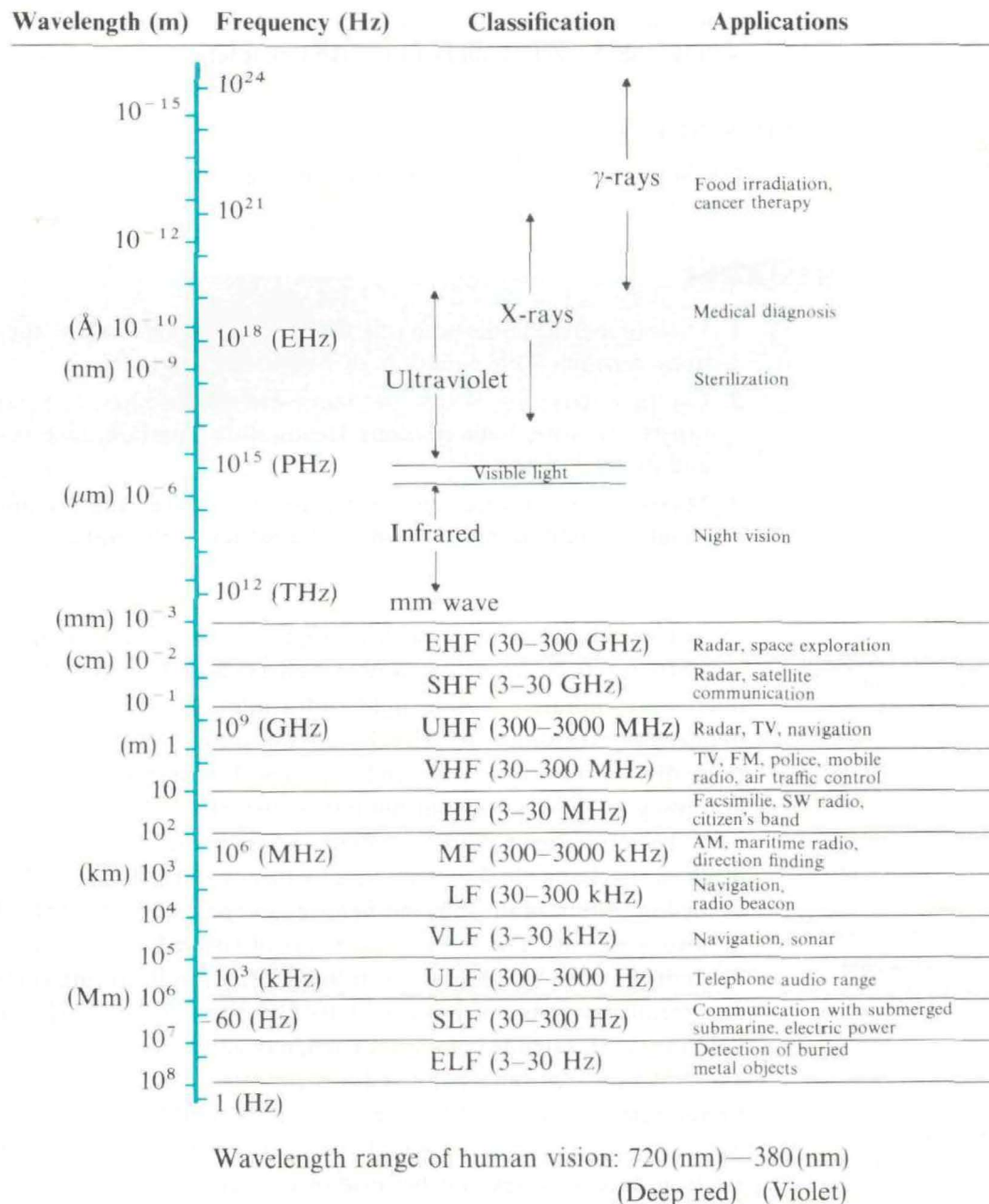


FIGURE 6-8 Spectrum of electromagnetic waves.

## REVIEW QUESTIONS

**Q.6-31** What is the wavelength range of visible light?

**Q.6-32** Why are frequencies below the VLF range rarely used for wireless transmission?

TABLE 6-4 RADAR BAND DESIGNATIONS FOR MICROWAVE FREQUENCIES

| Band | Frequency Range (GHz) | Wavelength Range (cm) |
|------|-----------------------|-----------------------|
| U    | 40–60                 | 0.75–0.50             |
| Ka   | 26.5–40               | 1.13–0.75             |
| K    | 18–26.5               | 1.67–1.13             |
| Ku   | 12.4–18               | 2.42–1.67             |
| X    | 8–12.4                | 3.75–2.42             |
| C    | 4–8                   | 7.5–3.75              |
| S    | 2–4                   | 15–7.5                |
| L    | 1–2                   | 30–15                 |

**REMARKS**

1. Electromagnetic waves of *all frequencies* propagate in a given medium with the same velocity,  $u_p = 1/\sqrt{\mu\epsilon}$ .
2. The operating frequency of microwave ovens is around 2.45 (GHz).

**SUMMARY**

In time-varying situations electric and magnetic fields are coupled and the postulates that we introduced in previous chapters for static fields no longer suffice. In this chapter we

- added a fundamental postulate for electromagnetic induction,
- introduced Faraday's law that relates quantitatively the emf induced in a circuit to the time rate of change of flux linkage,
- explained that the induced emf may be decomposed into two parts: a transformer emf and a motional (flux-cutting) emf,
- discussed the characteristics of ideal transformers,
- obtained a set of four Maxwell's equations (two divergence and two curl equations) that are consistent with the equation of continuity,
- considered the general boundary conditions for field vectors at the interface of contiguous regions of different constitutive parameters,
- expressed electric and magnetic field intensities in terms of a scalar electric potential function  $V$  and a vector magnetic potential function  $\mathbf{A}$ ,

- derived nonhomogeneous wave equations for  $V$  and  $\mathbf{A}$ ,
- introduced the concept of retarded potentials,
- converted wave equations into Helmholtz's equations for time-harmonic fields, and
- discussed the electromagnetic spectrum in source-free space.

## PROBLEMS

**P.6-1** Express the transformer emf induced in a stationary loop in terms of time-varying vector potential  $\mathbf{A}$ .

**P.6-2** The circuit in Fig. 6-9 is situated in a magnetic field

$$\mathbf{B} = \mathbf{a}_x 3 \cos(5\pi 10^7 t - \tfrac{1}{3}\pi y) \quad (\mu\text{T}).$$

Assuming  $R = 15(\Omega)$ , find the current  $i$ .

**P.6-3** A stationary rectangular conducting loop of width  $w$  and height  $h$  is situated near a very long wire carrying a current  $i_1$  as in Fig. 6-10. Assume  $i_1 = I_1 \sin \omega t$  and the self-inductance of the rectangular loop to be  $L$ . Find the induced current  $i_2$  in the loop.

**HINT:** Use phasors.

**P.6-4** In Fig. 6-10 assume a constant current  $i_1 = I_0$ , but that the rectangular loop moves away with a constant velocity  $\mathbf{u} = \mathbf{a}_y u_0$ . Determine  $i_2$  when the loop is at a position as shown.

**P.6-5** A 10(cm) by 10(cm) square conducting loop having a resistance  $R = 0.5(\Omega)$  rotates in a constant magnetic field  $\mathbf{B} = \mathbf{a}_y 0.04(\text{T})$  with an

FIGURE 6-9 A circuit in a time-varying magnetic field (Problem P6-2).

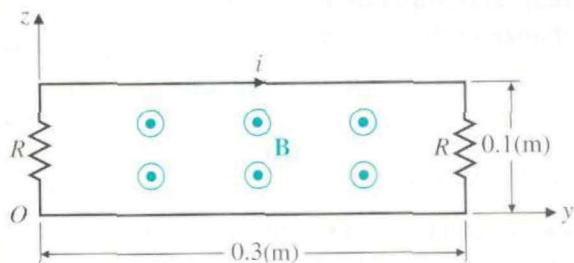
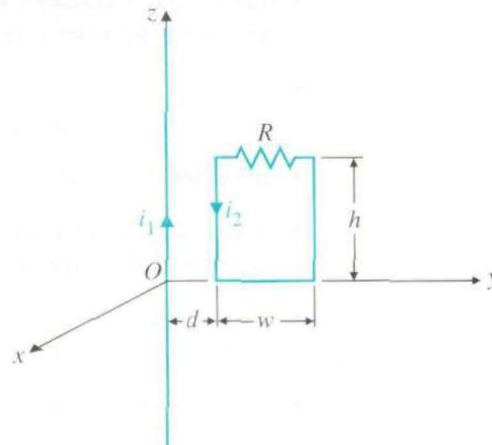


FIGURE 6-10 A rectangular loop near a long current-carrying wire (for Problems P.6-3 and P.6-4).



angular frequency  $\omega = 100\pi$  (rad/s) about one of its legs, as shown in Fig. 6-11. Assuming that the loop initially lies in the  $xz$ -plane, find the induced current  $i$

- if the self-inductance of the loop is neglected, and
- if the self-inductance of the loop is 3.5 (mH).

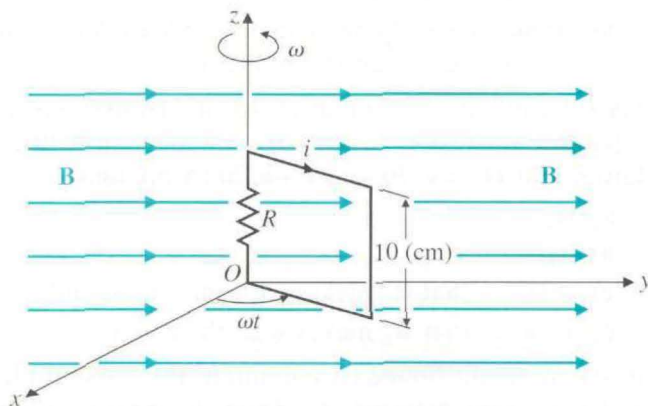


FIGURE 6-11 A rotating square loop in a constant magnetic field (Problem P.6-5).

**P.6-6** A conducting sliding bar oscillates over two parallel conducting rails in a sinusoidally varying magnetic field

$$\mathbf{B} = \mathbf{a}_z 5 \cos \omega t \quad (\text{mT}),$$

as shown in Fig. 6-12. The position of the sliding bar is given by  $x = 0.35(1 - \cos \omega t)$  (m), and the rails are terminated in a resistance  $R = 0.2$  ( $\Omega$ ). Find  $i$ .

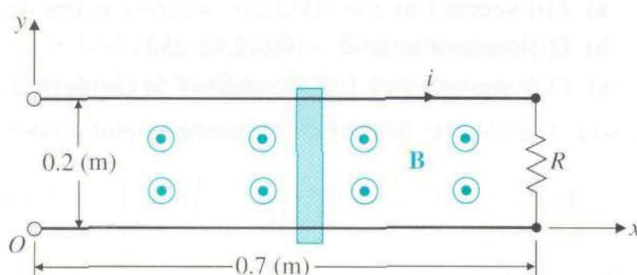


FIGURE 6-12 A conducting bar sliding over parallel rails in a time-varying magnetic field (Problem P.6-6).

**P.6-7** Determine the frequency at which a time-harmonic electric field intensity causes a conduction current density and a displacement current density of equal magnitude in

- seawater with  $\epsilon_r = 72$  and  $\sigma = 4$  (S/m), and
- moist soil with  $\epsilon_r = 2.5$  and  $\sigma = 10^{-3}$  (S/m).



**P.6-8** Calculations concerning the electromagnetic effect of currents in a good conductor usually neglect the displacement current even at microwave frequencies.

- Assuming  $\epsilon_r = 1$  and  $\sigma = 5.70 \times 10^7$  (S/m) for copper, compare the magnitude of the displacement current density with that of the conduction current density at 100(GHz).
- Write the governing differential equation for magnetic field intensity  $\mathbf{H}$  in a source-free good conductor.

**P.6-9** An infinite current sheet  $\mathbf{J} = \mathbf{a}_x 5$  (A/m) coinciding with the  $xy$ -plane separates air (region 1,  $z > 0$ ) from a medium with  $\mu_{r2} = 2$  (region 2,  $z < 0$ ). Given that  $\mathbf{H}_1 = \mathbf{a}_x 30 + \mathbf{a}_y 40 + \mathbf{a}_z 20$  (A/m), find

- $\mathbf{H}_2$ ,
- $\mathbf{B}_2$ ,
- angle  $\alpha_1$  that  $\mathbf{B}_1$  makes with the  $z$ -axis, and
- angle  $\alpha_2$  that  $\mathbf{B}_2$  makes with the  $z$ -axis.

**P.6-10** Write the boundary conditions that exist at the interface of free space and a magnetic material of infinite (an approximation) permeability.

**P.6-11** Prove by direct substitution that any twice differentiable function of  $(t - R\sqrt{\mu\epsilon})$  is a solution of the homogeneous wave equation, Eq. (6-61).

**P.6-12** Write the component scalar equations for the set of four Maxwell's equations (6-80a, b, c, and d):

- in Cartesian coordinates if the field phasors are functions of  $z$  only, and
- in spherical coordinates if the field phasors are functions of  $R$  only.

**P.6-13** Assuming  $E(z, t) = 50 \cos(2\pi 10^9 t - kz)$  (V/m) in air, sketch the following waveforms and label the abscissas:

- $E(t)$  versus  $t$  at  $z = 100.125\lambda$ , where  $\lambda$  is the wavelength,
- $E(t)$  versus  $t$  at  $z = -100.125\lambda$ , and
- $E(z)$  versus  $z$  at  $t = T/4$ , where  $T$  is the period of the wave.

**P.6-14** The electric field of an electromagnetic wave

$$\mathbf{E}(z, t) = \mathbf{a}_x E_0 \cos \left[ 10^8 \pi \left( t - \frac{z}{c} \right) + \psi \right] \quad (\text{V/m})$$

is the sum of

$$\mathbf{E}_1(z, t) = \mathbf{a}_x 0.03 \sin 10^8 \pi \left( t - \frac{z}{c} \right) \quad (\text{V/m})$$

and

$$\mathbf{E}_2(z, t) = \mathbf{a}_x 0.04 \cos \left[ 10^8 \pi \left( t - \frac{z}{c} \right) - \frac{\pi}{3} \right]. \quad (\text{V/m}).$$

Find  $E_0$  and  $\psi$ .

**P.6-15** The magnetic field intensity of an electromagnetic wave

$$\mathbf{H}(R, t) = \mathbf{a}_\phi H_0 \cos(\omega t - kR) \quad (\text{A/m})$$

is the sum of

$$\mathbf{H}_1(R, t) = \mathbf{a}_\phi 10^{-4} \sin(\omega t - kR) \quad (\text{A/m})$$

and

$$\mathbf{H}_2(z, t) = \mathbf{a}_\phi 2 \times 10^{-4} \cos(\omega t - kR + \alpha) \quad (\text{A/m}).$$

Find  $H_0$  and  $\alpha$ .

**P.6-16** Starting from the phasor Maxwell's equations (6-80a, b, c, and d) in simple media with time-harmonic charge and current distributions obtain

- the nonhomogeneous Helmholtz's equation for  $\mathbf{E}$ , and
- the nonhomogeneous Helmholtz's equation for  $\mathbf{H}$ .

**P.6-17** A short conducting wire carrying a time-harmonic current is a source of electromagnetic waves. Assuming that a uniform current  $i(t) = I_0 \cos \omega t$  flows in a very short wire  $d\ell$  placed along the  $z$ -axis,

- determine the phasor retarded vector potential  $\mathbf{A}$  at a distance  $R$  in spherical coordinates, and
- find the magnetic field intensity  $\mathbf{H}$  from  $\mathbf{A}$ .

**P.6-18** A 60-(MHz) electromagnetic wave exists in an air-dielectric coaxial cable having an inner conductor with radius  $a$  and an outer conductor with inner radius  $b$ . Assuming perfect conductors, and the phasor form of the electric field intensity to be

$$\mathbf{E} = \mathbf{a}_r \frac{E_0}{r} e^{-jkz} \quad (\text{V/m}), \quad a < r < b,$$

- find  $k$ ,
- find  $\mathbf{H}$  from the  $\nabla \times \mathbf{E}$  equation, and
- find the surface current densities on the inner and outer conductors.

**P.6-19** It is known that the electric field intensity of a spherical wave in free space is

$$\mathbf{E}(R, \theta; t) = \mathbf{a}_\theta \frac{10^{-3}}{R} \sin \theta \cos(2\pi 10^9 t - kR) \quad (\text{V/m}).$$

Determine the magnetic field intensity  $\mathbf{H}(R, \theta; t)$  and the value of  $k$ .

**P.6-20** Given that

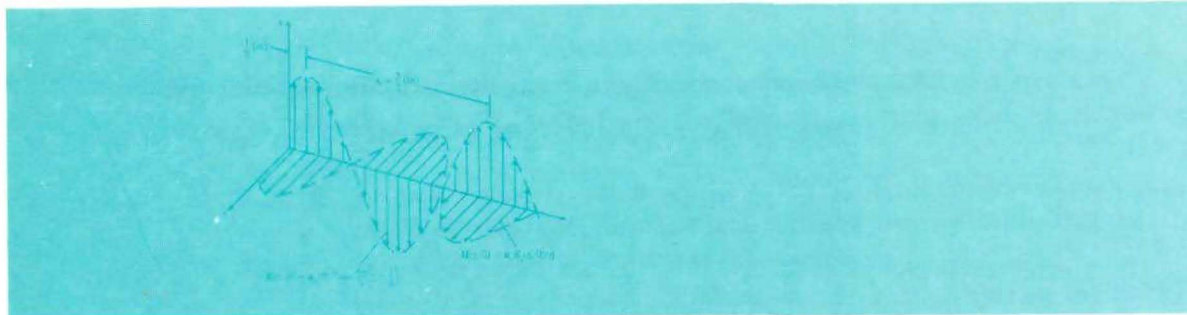
$$\mathbf{E}(x, z; t) = \mathbf{a}_y 0.1 \sin(10\pi x) \cos(6\pi 10^9 t - \beta z) \quad (\text{V/m})$$

in air, find  $\mathbf{H}(x, z; t)$  and  $\beta$ .

**P.6-21** Given that

$$\mathbf{H}(x, z; t) = \mathbf{a}_y 2 \cos(15\pi x) \sin(6\pi 10^9 t - \beta z) \quad (\text{A/m})$$

in air, find  $\mathbf{E}(x, z; t)$  and  $\beta$ .



## CHAPTER 7

**7-1 OVERVIEW** In Section 6-5 we showed that in a source-free nonconducting simple medium, Maxwell's equations can be combined to yield homogeneous vector wave equations in  $\mathbf{E}$  and in  $\mathbf{H}$ . These two equations, Eqs. (6-96) and (6-97), have exactly the same form. For example, the source-free wave equation for  $\mathbf{E}$  is

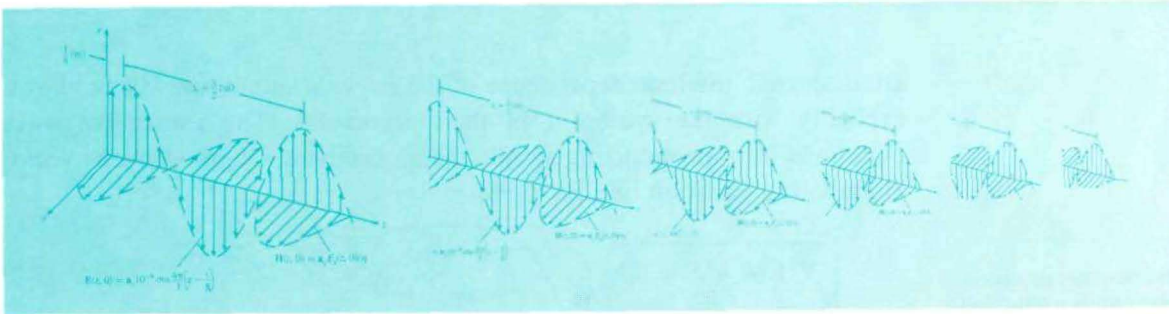
$$\nabla^2 \mathbf{E} - \frac{1}{u_p^2} \frac{\partial^2 \mathbf{E}}{\partial t^2} = 0. \quad (6-96)(7-1)$$

In Cartesian coordinates, Eq. (7-1) can be decomposed into three one-dimensional, homogeneous, scalar wave equations. Each component equation will be of the form of Eq. (6-61), whose solution represents a wave propagating in the medium with a velocity

$$u_p = \frac{1}{\sqrt{\mu\epsilon}}. \quad (6-63)(7-2)$$

Since time-varying  $\mathbf{E}$  and  $\mathbf{H}$  are coupled through Maxwell's curl equations (6-94a) and (6-94b), the consequence is an electromagnetic wave, which we use to explain electromagnetic action at a distance. The study of the behavior of waves that have a one-dimensional spatial dependence is the main concern of this chapter.

We begin the chapter with a study of the propagation of time-harmonic plane-wave fields in an unbounded homogeneous medium. Medium parameters such as intrinsic impedance, attenuation constant, and phase constant will be introduced. The meaning of *skin depth*, the depth of wave penetration into a



# Plane Electromagnetic Waves

good conductor, will be explained. Electromagnetic waves carry with them electromagnetic power. The concept of *Poynting vector*, a power flux density, will be discussed.

We will study the behavior of a plane wave incident obliquely on a plane boundary. The laws governing the reflection and refraction of plane waves as well as the conditions for no reflection and for total reflection will be examined.

Definition of  
uniform plane  
wave

A *uniform plane wave* is a particular solution of Maxwell's equations with  $\mathbf{E}$  assuming the same direction, same magnitude, and same phase in infinite planes perpendicular to the direction of propagation (similarly for  $\mathbf{H}$ ). Strictly speaking, a uniform plane wave does not exist in practice because a source infinite in extent would be required to create it, and practical wave sources are always finite in extent. But, if we are far enough away from a source, the *wavefront* (surface of constant phase) becomes almost spherical; and a very small portion of the surface of a giant sphere is very nearly a plane. The characteristics of uniform plane waves are particularly simple, and their study is of fundamental theoretical, as well as practical, importance.

Definition of  
Wavefront

## 7-2 PLANE WAVES IN LOSSLESS MEDIA

In this and future chapters we focus our attention on wave behavior in the sinusoidal steady state. Sinusoidal quantities will be represented by phasors without a subscript  $s$ , for simplicity. In situations where instantaneous time functions



are discussed, the time dependence of the relevant quantities will be shown explicitly with the symbol  $t$  in their arguments. The source-free wave equations in nonconducting simple media becomes a homogeneous vector Helmholtz's equation (see Eq. 6-98):

$$\nabla^2 \mathbf{E} + k^2 \mathbf{E} = 0, \quad (7-3)$$

Homogeneous vector  
Helmholtz's equation  
in  $\mathbf{E}$  in  
nonconducting  
simple media

where  $k$  is the wavenumber. In a medium characterized by  $\epsilon$  and  $\mu$ , we have, from Eq. (6-82a),

$$k = \omega \sqrt{\mu\epsilon} = \frac{\omega}{u_p} \quad (\text{rad/m}). \quad (7-4)$$

In Cartesian coordinates, Eq. (7-3) is equivalent to three scalar Helmholtz's equations, one each in the components  $E_x$ ,  $E_y$ , and  $E_z$ . Writing it for the component  $E_x$ , we have

$$\left( \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} + k^2 \right) E_x = 0. \quad (7-5)$$

Consider a uniform plane wave characterized by a uniform  $E_x$  (uniform magnitude and constant phase) over plane surfaces perpendicular to  $z$ ; that is,

$$\partial^2 E_x / \partial x^2 = 0 \quad \text{and} \quad \partial^2 E_x / \partial y^2 = 0.$$

Equation (7-5) simplifies to

$$\frac{d^2 E_x}{dz^2} + k^2 E_x = 0, \quad (7-6)$$

which is an ordinary differential equation because  $E_x$ , a phasor, depends only on  $z$ .

The solution of Eq. (7-6) is readily seen to be

$$\begin{aligned} E_x(z) &= E_x^+(z) + E_x^-(z) \\ &= E_0^+ e^{-jkz} + E_0^- e^{jkz}, \end{aligned} \quad (7-7)$$

where  $E_0^+$  and  $E_0^-$  are arbitrary constants that must be determined by boundary conditions.

Let us now examine the first phasor term on the right side of Eq. (7-7) and write

$$\mathbf{E}(z) = \mathbf{a}_x E_x^+(z) = \mathbf{a}_x E_0^+ e^{-jkz}. \quad (7-8)$$

For a cosine reference, the instantaneous expression for  $\mathbf{E}$  in Eq. (7-8) is

$$\begin{aligned} \mathbf{E}(z, t) &= \mathbf{a}_x E_x^+(z, t) = \mathbf{a}_x \Re [E_x^+(z) e^{j\omega t}] \\ &= \mathbf{a}_x \Re [E_0^+ e^{j(\omega t - kz)}] = \mathbf{a}_x E_0^+ \cos(\omega t - kz). \end{aligned} \quad (7-9)$$

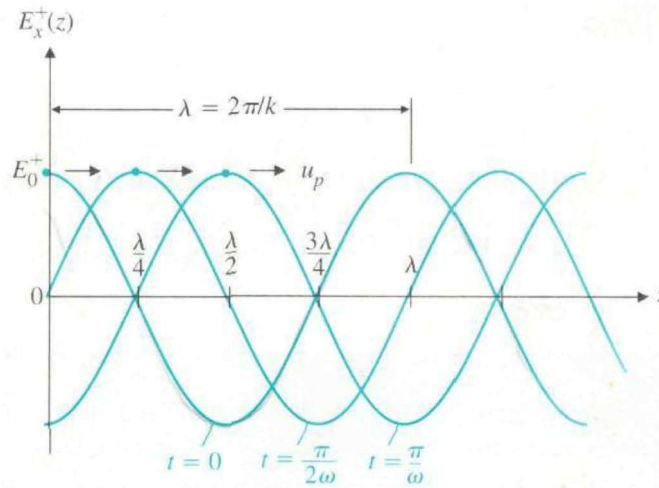


FIGURE 7-1 Wave traveling in positive  $z$  direction  $E_x^+(z, t) = E_0^+ \cos(\omega t - kz)$ , for several values of  $t$ .

#### Characteristics of a traveling wave

Equation (7-9) has been plotted in Fig. 7-1 for several values of  $t$ . At  $t = 0$ ,  $E_x^+(z, 0) = E_0^+ \cos kz$  is a cosine curve with an amplitude  $E_0^+$ . At successive times the curve effectively *travels in the positive  $z$  direction*. We have, then, a **traveling wave**. If we fix our attention on a particular point (a point of a particular phase) on the wave, we set  $\cos(\omega t - kz) = \text{a constant}$ , or

$$\omega t - kz = \text{A constant phase,}$$

from which we obtain

$$u_p = \frac{dz}{dt} = \frac{\omega}{k} = \frac{1}{\sqrt{\mu\epsilon}}. \quad (7-10)$$

#### Definition of phase velocity

Equation (7-10) assures us that the velocity of propagation of an equiphase front (the **phase velocity**) is equal to the velocity of light. Wavenumber  $k$  bears a definite relation to the wavelength.

$$k = \omega \sqrt{\mu\epsilon} = \frac{2\pi}{\lambda}, \quad (7-11)$$

as has been noted in Eqs. (6-82a and b).

We can see, without replottting, that the second phasor term on the right side of Eq. (7-7),  $E_0^- e^{jkz}$ , represents a cosinusoidal wave traveling in the  $-z$  direction with the same velocity  $u_p$ . If we are concerned only with the wave traveling in the  $+z$  direction, we set  $E_0^- = 0$ . However, if there are discontinuities in the medium, reflected waves traveling in the opposite direction must also be considered, as we will see later in this chapter.

The associated magnetic field  $\mathbf{H}$  can be found from the  $\nabla \times \mathbf{E}$  equation (6-80a):

$$\nabla \times \mathbf{E} = \begin{vmatrix} \mathbf{a}_x & \mathbf{a}_y & \mathbf{a}_z \\ 0 & 0 & \frac{\partial}{\partial z} \\ E_x^+(z) & 0 & 0 \end{vmatrix} = -j\omega\mu(\mathbf{a}_x H_x^+ + \mathbf{a}_y H_y^+ + \mathbf{a}_z H_z^+),$$

which leads to

$$H_x^+ = 0, \quad (7-12a)$$

$$H_y^+ = \frac{1}{-j\omega\mu} \frac{\partial E_x^+(z)}{\partial z}, \quad (7-12b)$$

$$H_z^+ = 0. \quad (7-12c)$$

Thus  $H_y^+$  is the only nonzero component of  $\mathbf{H}$  corresponding to the  $\mathbf{E}$  in Eq. (7-8), and since

$$\frac{\partial E_x^+(z)}{\partial z} = \frac{\partial}{\partial z} (E_0^+ e^{-jkz}) = -jkE_x^+(z),$$

Eq. (7-12b) yields

$$\begin{aligned} \mathbf{H} &= \mathbf{a}_y H_y^+(z) = \mathbf{a}_y \frac{k}{\omega\mu} E_x^+(z) \\ &= \mathbf{a}_y \frac{1}{\eta} E_x^+(z). \end{aligned} \quad (7-13)$$

We have introduced a new quantity,  $\eta$ , in Eq. (7-13):  $\eta = \omega\mu/k$ , or

$$\eta = \sqrt{\frac{\mu}{\epsilon}} \quad (\Omega), \quad (7-14)$$

which is called the *intrinsic impedance* of the medium. In air  $\eta_0 = \sqrt{\mu_0/\epsilon_0} = 120\pi = 377 \, (\Omega)$ .  $H_y^+(z)$  is in phase with  $E_x^+(z)$ , and we can write the instantaneous expression for  $\mathbf{H}$  as

$$\begin{aligned} \mathbf{H}(z, t) &= \mathbf{a}_y H_y^+(z, t) = \mathbf{a}_y \Re e [H_y^+(z) e^{j\omega t}] \\ &= \mathbf{a}_y \frac{E_0^+}{\eta} \cos(\omega t - kz). \end{aligned} \quad (7-15)$$

**Intrinsic impedance  
of medium**

### ■ EXERCISE 7.1

Starting from the phasor expression  $\mathbf{E} = \mathbf{a}_x E_x^-(z) = \mathbf{a}_x E_0^- e^{jkz}$  for a uniform plane wave traveling in the  $-z$  direction, determine the associated  $\mathbf{H} = \mathbf{a}_y H_y^-(z)$ . Evaluate the ratio  $E_x^-(z)/H_y^-(z)$ .

ANS.  $E_x/H_y^- = -\eta$ .

## REVIEW QUESTIONS

**Q.7-1** Define *uniform plane wave*.

**Q.7-2** What is a *wavefront*?

**Q.7-3** What is a *traveling wave*?

**Q.7-4** Define *phase velocity*.

**Q.7-5** Define *intrinsic impedance* of a medium. What is the value of the intrinsic impedance of free space?

## REMARKS

1. The phase velocity of an electromagnetic wave in a lossless medium is independent of its frequency or its direction of travel.
2. The ratio of the magnitudes of  $\mathbf{E}$  and  $\mathbf{H}$  for a uniform plane wave is the intrinsic impedance of the medium.
3. The direction of the  $\mathbf{E}$ -field crosses into the direction of the  $\mathbf{H}$ -field gives the direction of wave propagation; the three directions are mutually perpendicular.

## EXAMPLE 7-1

A uniform plane wave with  $\mathbf{E} = \mathbf{a}_x E_x$  propagates in a lossless simple medium ( $\epsilon_r = 4$ ,  $\mu_r = 1$ ,  $\sigma = 0$ ) in the  $+z$ -direction. Assume that  $E_x$  is sinusoidal with a frequency 100 (MHz) and has a maximum value of  $+10^{-4}$  (V/m) at  $t = 0$  and  $z = \frac{1}{8}$  (m).

- a) Write the instantaneous expression for  $\mathbf{E}$  for any  $t$  and  $z$ .
- b) Write the instantaneous expression for  $\mathbf{H}$ .
- c) Determine the locations where  $E_x$  is a positive maximum when  $t = 10^{-8}$  (s).

## SOLUTION

First we find  $k$ :

$$\begin{aligned} k &= \omega \sqrt{\mu \epsilon} = \frac{\omega}{c} \sqrt{\mu_r \epsilon_r} \\ &= \frac{2\pi 10^8}{3 \times 10^8} \sqrt{4} = \frac{4\pi}{3} \quad (\text{rad/m}). \end{aligned}$$

- a) Using  $\cos \omega t$  as the reference, we find the instantaneous expression for  $\mathbf{E}$  to be

$$\mathbf{E}(z, t) = \mathbf{a}_x E_x = \mathbf{a}_x 10^{-4} \cos(2\pi 10^8 t - kz + \psi).$$



Since  $E_x$  equals  $+10^{-4}$  when the argument of the cosine function equals zero—that is, when

$$2\pi 10^8 t - kz + \psi = 0,$$

we have, at  $t = 0$  and  $z = \frac{1}{8}$ ,

$$\psi = kz = \left(\frac{4\pi}{3}\right)\left(\frac{1}{8}\right) = \frac{\pi}{6} \quad (\text{rad}).$$

Thus,

$$\begin{aligned} E(z, t) &= a_x 10^{-4} \cos\left(2\pi 10^8 t - \frac{4\pi}{3} z + \frac{\pi}{6}\right) \\ &= a_x 10^{-4} \cos\left[2\pi 10^8 t - \frac{4\pi}{3}\left(z - \frac{1}{8}\right)\right] \quad (\text{V/m}). \end{aligned}$$

This expression shows a shift of  $\frac{1}{8}$  (m) in the  $+z$ -direction and could have been written down directly from the statement of the problem.

- b) The instantaneous expression for  $\mathbf{H}$  is

$$\mathbf{H} = \mathbf{a}_y H_y = \mathbf{a}_y \frac{E_x}{\eta},$$

where

$$\eta = \sqrt{\frac{\mu}{\epsilon}} = \frac{\eta_0}{\sqrt{\epsilon_r}} = 60\pi \quad (\Omega).$$

Hence,

$$\mathbf{H}(z, t) = \mathbf{a}_y \frac{10^{-4}}{60\pi} \cos\left[2\pi 10^8 t - \frac{4\pi}{3}\left(z - \frac{1}{8}\right)\right] \quad (\text{A/m}).$$

- c) At  $t = 10^{-8}$ , we equate the argument of the cosine function to  $\pm 2n\pi$  in order to make  $E_x$  a positive maximum:

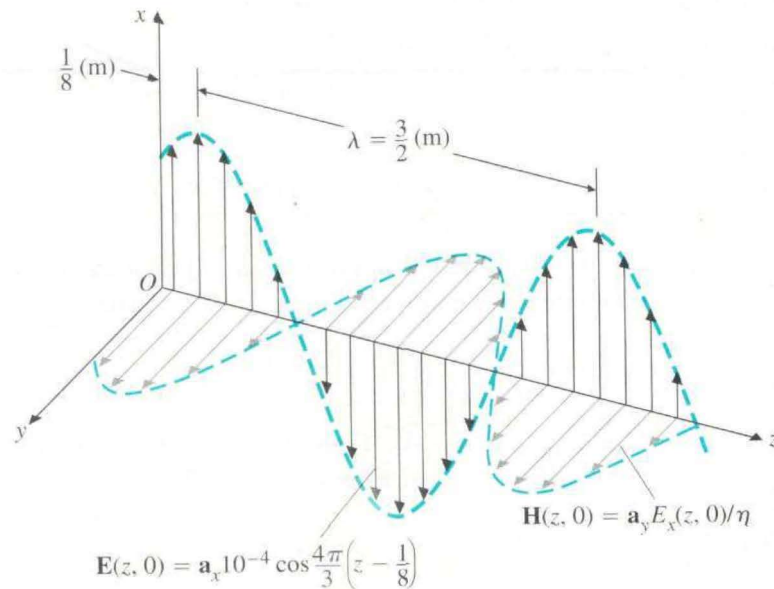
$$2\pi 10^8 (10^{-8}) - \frac{4\pi}{3}\left(z_m - \frac{1}{8}\right) = \pm 2n\pi,$$

from which we get

$$z_m = \frac{13}{8} \pm \frac{3}{2}n \quad (\text{m}), \quad n = 0, 1, 2, \dots; \quad z_m > 0.$$

Examining this result more closely, we note that the wavelength in the given medium is

$$\lambda = \frac{2\pi}{k} = \frac{3}{2} \quad (\text{m}).$$

FIGURE 7-2  $\mathbf{E}$  and  $\mathbf{H}$  fields of a uniform plane wave at  $t = 0$  (Example 7-1).

Hence the positive maximum value of  $E_x$  occurs at

$$z_m = \frac{13}{8} \pm n\lambda \quad (\text{m}).$$

The  $\mathbf{E}$  and  $\mathbf{H}$  fields are shown in Fig. 7-2 as functions of  $z$  for the reference time  $t = 0$ .

### 7-2.1 DOPPLER EFFECT

#### Meaning of Doppler effect

When there is relative motion between a time-harmonic source and a receiver, the frequency of the wave detected by the receiver tends to be different from that emitted by the source. This phenomenon is known as **Doppler effect**.<sup>†</sup> The Doppler effect manifests itself in acoustics as well as in electromagnetics. Perhaps you have experienced the changes in the pitch of a fast-moving locomotive whistle. In the following we give an explanation of the Doppler effect.

Let us assume that the source (transmitter)  $T$  of a time-harmonic wave of a frequency  $f$  moves with a velocity  $\mathbf{u}$  at an angle  $\theta$  relative to the direct

<sup>†</sup>C. Doppler (1803–1853).



FIGURE 7-3 Illustrating the Doppler effect.

line to a stationary receiver  $R$ , as illustrated in Fig. 7-3(a). The electromagnetic wave emitted by  $T$  in air at a reference time  $t = 0$  will reach  $R$  at

$$t_1 = \frac{r_0}{c}. \quad (7-16)$$

At a later time  $t = \Delta t$ ,  $T$  has moved to the new position  $T'$ , and the wave emitted by  $T'$  at that time will reach  $R$  at

$$\begin{aligned} t_2 &= \Delta t + \frac{r'}{c} \\ &= \Delta t + \frac{1}{c} [r_0^2 - 2r_0(u \Delta t) \cos \theta + (u \Delta t)^2]^{1/2}. \end{aligned} \quad (7-17)$$

If  $(u \Delta t)^2 \ll r_0^2$ , Eq. (7-17) becomes

$$t_2 \cong \Delta t + \frac{r_0}{c} \left( 1 - \frac{u \Delta t}{r_0} \cos \theta \right).$$

Hence the time elapsed at  $R$ ,  $\Delta t'$ , corresponding to  $\Delta t$  at  $T$  is

$$\begin{aligned} \Delta t' &= t_2 - t_1 \\ &= \Delta t \left( 1 - \frac{u}{c} \cos \theta \right), \end{aligned} \quad (7-18)$$

which is not equal to  $\Delta t$ .

If  $\Delta t$  represents a period of the time-harmonic source—that is, if  $\Delta t = 1/f$ —then the frequency of the received wave at  $R$  is

$$\begin{aligned} f' &= \frac{1}{\Delta t'} = \frac{f}{\left( 1 - \frac{u}{c} \cos \theta \right)} \\ &\cong f \left( 1 + \frac{u}{c} \cos \theta \right) \end{aligned} \quad (7-19)$$

for the usual case of  $(u/c)^2 \ll 1$ . Equation (7-19) clearly indicates that the frequency perceived at  $R$  is higher than the transmitted frequency when  $T$  moves toward  $R$ . Conversely, the perceived frequency is lower than the transmitted frequency when  $T$  moves away from  $R$ . Similar results are obtained if  $R$  moves and  $T$  is stationary.

Principle of operation of radar speed monitor

Red shift phenomenon in astronomy

The Doppler effect is the basis of operation of the (Doppler) radar used by police to check the speed of a moving vehicle. The frequency shift of the received wave reflected by a moving vehicle is proportional to the speed of the vehicle and can be detected and displayed on a hand-held unit. The Doppler effect is also the cause of the so-called **red shift** of the light spectrum emitted by a receding distant star in astronomy. As the star *moves away* at a high speed from an observer on earth, the received frequency shifts toward the lower frequency (red) end of the spectrum.

### ■ EXERCISE 7.2

A train traveling at 130 (km/hr) toward an observer at an angle  $20^\circ$  from the line of sight. It gives out a 800 (Hz) whistle. What is the frequency perceived by the observer? (The velocity of sound is approximately 340 m/s.)

ANS. 880 (Hz).

### REVIEW QUESTION

Q.7-6 What is Doppler effect?

### REMARKS

The frequency of a wave detected by a receiver is higher (lower) than that emitted by a transmitter if the transmitter moves toward (away from) the receiver.

## 7-2.2 TRANSVERSE ELECTROMAGNETIC WAVES

TEM wave

We have seen that a uniform plane wave characterized by  $\mathbf{E} = \mathbf{a}_x E_x$  propagating in the  $+z$ -direction has associated with it a magnetic field  $\mathbf{H} = \mathbf{a}_y H_y$ . Thus  $\mathbf{E}$  and  $\mathbf{H}$  are perpendicular to each other, and both are *transverse* to the direction of propagation. It is a particular case of a **transverse electromagnetic (TEM) wave**. We now examine the propagation of a uniform plane wave along an arbitrary direction that does not necessarily coincide with a coordinate axis.

Instead of the  $\mathbf{E}(z)$  in Eq. (7-8), let us consider

$$\mathbf{E}(x, z) = \mathbf{a}_y E_0 e^{-jk_x x - jk_z z}, \quad (7-20)$$

which represents the  $y$ -directed electric intensity of a uniform plane wave propagating in both  $+x$  and  $+z$  directions. If we define a **wavenumber vector**,  $\mathbf{k}$ , as

$$\mathbf{k} = \mathbf{a}_x k_x + \mathbf{a}_z k_z = \mathbf{a}_k k, \quad (7-21)$$

and a radius vector  $\mathbf{R}$  from the origin to an arbitrary point

$$\mathbf{R} = \mathbf{a}_x x + \mathbf{a}_y y + \mathbf{a}_z z, \quad (7-22)$$

Expression of wavenumber vector in Cartesian coordinates



Eq. (7-20) can be written succinctly as

$$\mathbf{E} = \mathbf{a}_y E_0 e^{-j\mathbf{k} \cdot \mathbf{R}} = \mathbf{a}_y E_0 e^{-jk\mathbf{a}_k \cdot \mathbf{R}}. \quad (7-23)$$

The situation is illustrated in Fig. 7-4. The relation

$$\mathbf{a}_k \cdot \mathbf{R} = \text{Length } \overline{OP} \quad (\text{a constant})$$

is the equation of a plane (locus of the endpoints of radius vector  $\mathbf{R}$ ) normal to  $\mathbf{a}_k$ , the direction of propagation, and it is a plane of constant phase and uniform amplitude.

The magnetic field  $\mathbf{H}$  associated with the electric field in Eq. (7-23) is, from Eq. (6-80a),

$$\mathbf{H} = -\frac{1}{j\omega\mu} \nabla \times \mathbf{E} = \frac{E_0}{\omega\mu} (-\mathbf{a}_x k_z + \mathbf{a}_z k_x) e^{-jk_x x - jk_z z}. \quad (7-24)$$

Equation (7-24) can be put in a more general form:

$$\mathbf{H} = \frac{k}{\omega\mu} \mathbf{a}_k \times \mathbf{E} = \frac{1}{\eta} \mathbf{a}_k \times \mathbf{E}. \quad (7-25)$$

Finding  $\mathbf{H}$  from  $\mathbf{E}$  of  
a uniform plane  
wave

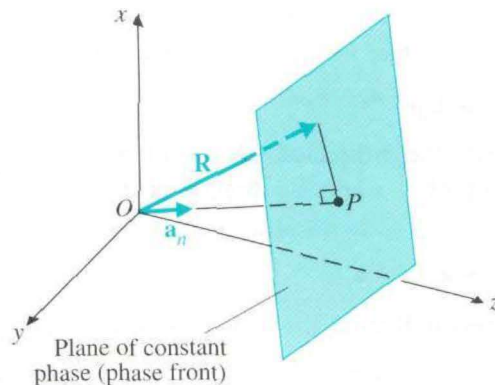
Thus, if  $\mathbf{E}$  of a uniform plane wave propagating in a given direction is known, the associated  $\mathbf{H}$  can be easily found from Eq. (7-25).

### EXERCISE 7.3

- Write the phasor expression of the  $z$ -directed electric field of a uniform plane wave in air having an amplitude  $E_0$ , frequency  $f$ , and traveling in the  $-y$  direction.
- Write the expression of the associated magnetic field.

ANS. (a)  $\mathbf{E} = \mathbf{a}_z E_0 e^{j2\pi f y/c}$ .

FIGURE 7-4 Radius vector and wave normal to a phase front of a uniform plane wave.



## 7-2.3 POLARIZATION OF PLANE WAVES

Meaning of  
polarization of a  
uniform wave

The **polarization** of a uniform plane wave describes the time-varying behavior of the electric field intensity vector at a given point in space. For example, if the  $\mathbf{E}$  vector of a plane wave is fixed in the  $x$  direction ( $\mathbf{E} = \mathbf{a}_x E_x$ , where  $E_x$  may be positive or negative), the wave is said to be **linearly polarized** in the  $x$ -direction. A separate description of magnetic-field behavior is not necessary, inasmuch as the direction of  $\mathbf{H}$  is definitely related to that of  $\mathbf{E}$ .

In some cases the direction of  $\mathbf{E}$  of a plane wave at a given point may change with time. Consider the superposition of two linearly polarized waves: one polarized in the  $x$ -direction, and the other polarized in the  $y$ -direction and lagging  $90^\circ$  (or  $\pi/2$  rad) in time phase. In phasor notation we have

$$\begin{aligned}\mathbf{E}(z) &= \mathbf{a}_x E_1(z) + \mathbf{a}_y E_2(z) \\ &= \mathbf{a}_x E_{10} e^{-jkz} - \mathbf{a}_y j E_{20} e^{-jkz},\end{aligned}\quad (7-26)$$

where  $E_{10}$  and  $E_{20}$  are real numbers denoting the amplitudes of the two linearly polarized waves. The instantaneous expression for  $\mathbf{E}$  is

$$\begin{aligned}\mathbf{E}(z, t) &= \Re e \{ [\mathbf{a}_x E_1(z) + \mathbf{a}_y E_2(z)] e^{j\omega t} \} \\ &= \mathbf{a}_x E_{10} \cos(\omega t - kz) + \mathbf{a}_y E_{20} \cos\left(\omega t - kz - \frac{\pi}{2}\right).\end{aligned}$$

In examining the direction change of  $\mathbf{E}$  at a given point as  $t$  changes, it is convenient to set  $z = 0$ . We have

$$\begin{aligned}\mathbf{E}(0, t) &= \mathbf{a}_x E_1(0, t) + \mathbf{a}_y E_2(0, t) \\ &= \mathbf{a}_x E_{10} \cos \omega t + \mathbf{a}_y E_{20} \sin \omega t.\end{aligned}\quad (7-27)$$

As  $\omega t$  increases from 0 through  $\pi/2$ ,  $\pi$ , and  $3\pi/2$ —completing the cycle at  $2\pi$ —the tip of the vector  $\mathbf{E}(0, t)$  will traverse an elliptical locus in the counterclockwise direction. Analytically, we have

$$\cos \omega t = \frac{E_1(0, t)}{E_{10}}$$

and

$$\begin{aligned}\sin \omega t &= \frac{E_2(0, t)}{E_{20}} \\ &= \sqrt{1 - \cos^2 \omega t} = \sqrt{1 - \left[ \frac{E_1(0, t)}{E_{10}} \right]^2},\end{aligned}$$

which leads to the following equation for an ellipse:

$$\left[ \frac{E_2(0, t)}{E_{20}} \right]^2 + \left[ \frac{E_1(0, t)}{E_{10}} \right]^2 = 1. \quad (7-28)$$

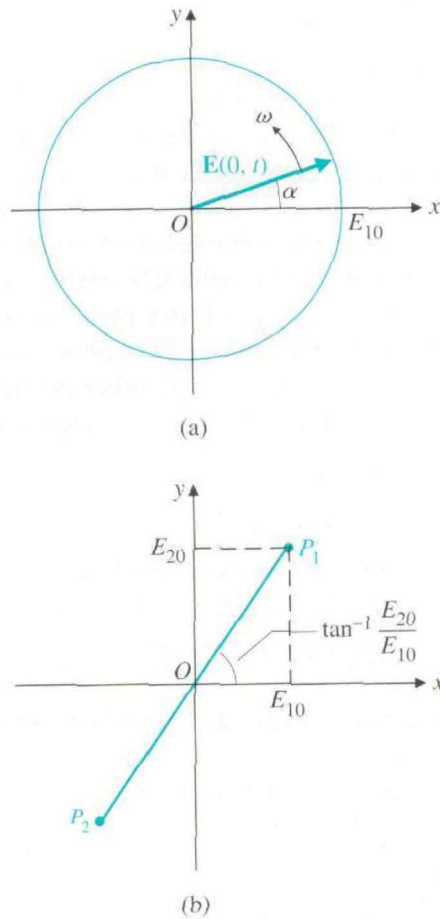


FIGURE 7-5 Polarization diagrams for sum of two linearly polarized waves in space quadrature at  $z = 0$ : (a) circular polarization,  $E(0, t) = E_{10}(\mathbf{a}_x \cos \omega t + \mathbf{a}_y \sin \omega t)$ ; (b) linear polarization,  $\mathbf{E}(0, t) = (\mathbf{a}_x E_{10} + \mathbf{a}_y E_{20}) \cos \omega t$ .

Meaning of  
elliptically and  
circularly polarized  
waves

Hence  $\mathbf{E}$ , which is the sum of two linearly polarized waves in both space and time quadrature, is **elliptically polarized** if  $E_{20} \neq E_{10}$ , and is **circularly polarized** if  $E_{20} = E_{10}$ . A typical polarization circle is shown in Fig. 7-5(a).

When  $E_{20} = E_{10}$ , the instantaneous angle  $\alpha$  that  $\mathbf{E}$  makes with the  $x$ -axis at  $z = 0$  is

$$\alpha = \tan^{-1} \frac{E_2(0, t)}{E_1(0, t)} = \omega t, \quad (7-29)$$

which indicates that  $\mathbf{E}$  rotates at a uniform rate with an angular velocity  $\omega$  in a *counterclockwise* direction. When the fingers of the right hand follow the direction of the rotation of  $\mathbf{E}$ , the thumb points to the direction of

Right-hand (or positive) circularly polarized wave

propagation of the wave. This is a **right-hand** or **positive circularly polarized wave**.

If we start with an  $E_2(z)$ , which *leads*  $E_1(z)$  by  $90^\circ$  ( $\pi/2$  rad) in time phase, Eqs. (7-26) and (7-27) will be, respectively,

$$\mathbf{E}(z) = \mathbf{a}_x E_{10} e^{-jkz} + \mathbf{a}_y j E_{20} e^{-jkz} \quad (7-30)$$

and

$$\mathbf{E}(0, t) = \mathbf{a}_x E_{10} \cos \omega t - \mathbf{a}_y E_{20} \sin \omega t. \quad (7-31)$$

Comparing Eq. (7-31) with Eq. (7-27), we see that  $\mathbf{E}$  will still be elliptically polarized. If  $E_{20} = E_{10}$ ,  $\mathbf{E}$  will be circularly polarized, and its angle measured from the  $x$ -axis at  $z = 0$  will now be  $-\omega t$ , indicating that  $\mathbf{E}$  will rotate with an angular velocity  $\omega$  in a *clockwise* direction; this is a **left-hand** or **negative circularly polarized wave**.

If  $E_2(z)$  and  $E_1(z)$  are in space quadrature but in time phase, the instantaneous expression for  $\mathbf{E}$  at  $z = 0$  is

$$\mathbf{E}(0, t) = (\mathbf{a}_x E_{10} + \mathbf{a}_y E_{20}) \cos \omega t. \quad (7-32)$$

The tip of the  $\mathbf{E}(0, t)$  will be at the point  $P_1$  in Fig. 7-5(b) when  $\omega t = 0$ . Its magnitude will decrease toward zero as  $\omega t$  increases toward  $\pi/2$ . After that,  $\mathbf{E}(0, t)$  starts to increase again, in the opposite direction, toward the point  $P_2$  where  $\omega t = \pi$ . We say that the sum  $\mathbf{E}$  is linearly polarized along a line that makes an angle  $\tan^{-1}(E_{20}/E_{10})$  with the  $x$ -axis.

In the general case,  $E_2(z)$  and  $E_1(z)$ , which are in space quadrature, can have unequal amplitudes ( $E_{20} \neq E_{10}$ ) and can differ in phase by an arbitrary amount (not zero or an integral multiple of  $\pi/2$ ). Their sum  $\mathbf{E}$  will be elliptically polarized.

Left-hand (or negative) circularly polarized wave

Linearly polarized wave

### EXAMPLE 7-2

Prove that a linearly polarized plane wave can be resolved into a right-hand circularly polarized wave and a left-hand circularly polarized wave of equal amplitude.

#### SOLUTION

Consider a linearly polarized plane wave propagating in the  $+z$ -direction. We can assume, with no loss of generality, that  $\mathbf{E}$  is polarized in the  $x$ -direction. In phasor notation we have

$$\mathbf{E}(z) = \mathbf{a}_x E_0 e^{-jkz}. \quad (7-33)$$

But this can be written as

$$\mathbf{E}(z) = \mathbf{E}_{rc}(z) + \mathbf{E}_{lc}(z), \quad (7-34)$$



where

$$\mathbf{E}_{rc}(z) = \frac{E_0}{2} (\mathbf{a}_x - j\mathbf{a}_y) e^{-jkz} \quad (7-34a)$$

and

$$\mathbf{E}_{lc}(z) = \frac{E_0}{2} (\mathbf{a}_x + j\mathbf{a}_y) e^{-jkz}. \quad (7-34b)$$

A linearly polarized plane wave can be resolved into two circularly polarized waves of equal magnitude.

From previous discussions we recognize that  $\mathbf{E}_{rc}(z)$  in Eq. (7-34a) and  $\mathbf{E}_{lc}(z)$  in Eq. (7-34b) represent right-hand and left-hand circularly polarized waves, respectively, each having an amplitude  $E_0/2$ . The statement of this problem is therefore proved. The converse statement that the sum of two oppositely rotating circularly polarized waves of equal amplitude is a linearly polarized wave is, of course, also true.

#### ■ EXERCISE 7.4

Describe the polarization of a wave whose electric intensity is represented by  $\mathbf{E}(x, t) = (\mathbf{a}_y E_{10} - \mathbf{a}_z E_{20}) \sin(\omega t - kx)$ .

### REVIEW QUESTIONS

**Q.7-7** What is a TEM wave?

**Q.7-8** Is the intrinsic impedance of a lossless medium a function of frequency? Explain.

**Q.7-9** What is meant by the *polarization* of a wave? When is a wave linearly polarized? Circularly polarized?

**Q.7-10** Two orthogonal linearly polarized waves of the same frequency are combined. State the conditions under which the resultant will be (a) another linearly polarized wave, (b) a circularly polarized wave, and (c) an elliptically polarized wave.

### REMARKS

1. For a TEM wave in an unbounded lossless medium: (a)  $\mathbf{E}$  and  $\mathbf{H}$  are in phase, and (b)  $|\mathbf{E}| = \eta|\mathbf{H}|$ .
2. The  $\mathbf{E}$ -field from AM broadcast stations is linearly polarized with the  $\mathbf{E}$ -field perpendicular to the ground. For maximum reception the receiving antenna should be in a vertical position.
3. The  $\mathbf{E}$ -field of television signals is linearly polarized in the horizontal direction. Note the horizontal position of TV receiving antennas on roof tops.
4. The wave radiated by FM broadcast stations are generally circularly polarized.

### 7-3 PLANE WAVES IN LOSSY MEDIA

So far we have considered wave propagation in source-free ( $\rho_v = 0, \mathbf{J} = 0$ ) lossless simple media. If a medium is conducting ( $\sigma \neq 0$ ), a current  $\mathbf{J} = \sigma \mathbf{E}$  will flow because of the existence of  $\mathbf{E}$ . The time-harmonic  $\nabla \times \mathbf{H}$  equation (6-80b) should then be changed to

$$\begin{aligned}\nabla \times \mathbf{H} &= (\sigma + j\omega\epsilon)\mathbf{E} = j\omega\left(\epsilon + \frac{\sigma}{j\omega}\right)\mathbf{E} \\ &= j\omega\epsilon_c\mathbf{E}\end{aligned}\quad (7-35)$$

with

$$\epsilon_c = \epsilon - j\frac{\sigma}{\omega} \quad (\text{F/m}). \quad (7-36)$$

Complex permittivity  
of lossy media

The other three equations, Eqs. (6-80a, c, and d), are unchanged. Hence, all the previous equations for nonconducting media will apply to conducting media if  $\epsilon$  is replaced by the **complex permittivity**  $\epsilon_c$  in Eq. (7-36).

As we discussed in Subsection 3-6.2, when an external time-varying electric field is applied to material bodies, small displacements of bound charges result, giving rise to a volume density of polarization. This polarization vector will vary with the same frequency as that of the applied field. As the frequency increases, the inertia of the charged particles tends to prevent the particle displacements from keeping in phase with the field changes, leading to a frictional damping mechanism that causes power loss because work must be done to overcome the damping forces. This phenomenon of out-of-phase polarization can be characterized by a complex electric susceptibility and hence a complex permittivity. If, in addition, the material body or medium has an appreciable amount of free charge carriers such as the electrons in a conductor, the electrons and holes in a semiconductor, or the ions in an electrolyte, there will also be ohmic losses. In treating such media it is customary to include the effects of both the damping and the ohmic losses in the imaginary part of a complex permittivity  $\epsilon_c$ :

$$\epsilon_c = \epsilon' - j\epsilon'' \quad (\text{F/m}), \quad (7-37)$$

where both  $\epsilon'$  and  $\epsilon''$  may be functions of frequency. Alternatively, we may define an equivalent conductivity representing all losses and write

$$\sigma = \omega\epsilon'' \quad (\text{S/m}). \quad (7-38)$$

Combination of Eqs. (7-37) and (7-38) gives Eq. (7-36).

The ratio  $\epsilon''/\epsilon'$  is called a **loss tangent** because it is a measure of the power loss in the medium:

$$\tan \delta_c = \frac{\epsilon''}{\epsilon'} \cong \frac{\sigma}{\omega\epsilon}. \quad (7-39)$$

The quantity  $\delta_c$  in Eq. (7-39) may be called the **loss angle**.

Loss tangent of a  
lossy medium

**Difference between  
a good conductor  
and a good insulator**

A medium is said to be a **good conductor** if  $\sigma \gg \omega\epsilon$ , and a **good insulator** if  $\omega\epsilon \gg \sigma$ . Thus, a material may be a good conductor at low frequencies but may have the properties of a lossy dielectric at very high frequencies. For example, a moist ground has a dielectric constant  $\epsilon_r$  and a conductivity  $\sigma$  that are in the neighborhood of 10 and  $10^{-2}$  (S/m), respectively. The loss tangent  $\sigma/\omega\epsilon$  of the moist ground then equals  $1.8 \times 10^4$  at 1 (kHz), making it a relatively good conductor. At 10 (GHz),  $\sigma/\omega\epsilon$  becomes  $1.8 \times 10^{-3}$ , and the moist ground behaves more like an insulator.

**EXAMPLE 7-3**

A sinusoidal electric intensity of amplitude 250 (V/m) and frequency 1 (GHz) exists in a lossy dielectric medium that has a relative permittivity of 2.5 and a loss tangent of 0.001. Find the average power dissipated in the medium per cubic meter.

**SOLUTION**

First we must find the effective conductivity of the lossy medium:

$$\begin{aligned}\tan \delta_c &= 0.001 = \frac{\sigma}{\omega\epsilon_0\epsilon_r}, \\ \sigma &= 0.001(2\pi 10^9) \left( \frac{10^{-9}}{36\pi} \right) (2.5) \\ &= 1.39 \times 10^{-4} \text{ (S/m)}.\end{aligned}$$

The average power dissipated per unit volume is

$$\begin{aligned}p &= \frac{1}{2}JE = \frac{1}{2}\sigma E^2 \\ &= \frac{1}{2} \times (1.39 \times 10^{-4}) \times 250^2 = 4.34 \quad (\text{W/m}^3).\end{aligned}$$

**EXERCISE 7.5**

A microwave oven cooks food by irradiating the food with microwave power generated by a magnetron. Assuming the dielectric constant of a beef steak to be 40 with a loss tangent of 0.35 at an operating frequency 2.45 (GHz), calculate the average power per cubic meter. (Neglect *skin effect*, which will be discussed in Subsection 7-3.2.)

ANS. 59.6 (kW/m<sup>3</sup>).

In view of the above discussion, the study of the time-harmonic behavior in lossy media can proceed from Eq. (7-3) by simply replacing the real  $k$  by a complex wavenumber  $k_c$ :

$$k_c = \omega\sqrt{\mu\epsilon_c}. \quad (7-40)$$

We need to examine the solution of the following homogeneous Helmholtz's



equation:

$$\nabla^2 \mathbf{E} + k_c^2 \mathbf{E} = 0. \quad (7-41)$$

In an effort to conform with the conventional notation used in transmission-line theory, it is customary to define a **propagation constant**,  $\gamma$ , such that

$$\gamma = jk_c = j\omega\sqrt{\mu\epsilon_c} \quad (\text{m}^{-1}). \quad (7-42)$$

Relation between  
propagation constant  
and wavenumber

Since  $\gamma$  is complex, we write, with the help of Eq. (7-36),

$$\gamma = \alpha + j\beta = j\omega\sqrt{\mu\epsilon} \left( 1 + \frac{\sigma}{j\omega\epsilon} \right)^{1/2}, \quad (7-43)$$

or, from Eq. (7-37),

$$\gamma = \alpha + j\beta = j\omega\sqrt{\mu\epsilon'} \left( 1 - j\frac{\epsilon''}{\epsilon'} \right)^{1/2}, \quad (7-44)$$

where  $\alpha$  and  $\beta$  are the real and imaginary parts of  $\gamma$ , respectively. Their physical significance will be explained presently. For a lossless medium,  $\sigma = 0$  ( $\epsilon'' = 0$ ,  $\epsilon = \epsilon'$ ),  $\alpha = 0$ , and  $\beta = k = \omega\sqrt{\mu\epsilon}$ .

With Eq. (7-42), Eq. (7-41) becomes

$$\nabla^2 \mathbf{E} - \gamma^2 \mathbf{E} = 0. \quad (7-45a)$$

For a uniform plane wave propagating in the  $+z$  direction and characterized by  $\mathbf{E} = \mathbf{a}_x E_x$  and  $\mathbf{H} = \mathbf{a}_y H_y$ , Eq. (7-45a) reduces to

$$\frac{d^2 E_x}{dz^2} = \gamma^2 E_x. \quad (7-45b)$$

The solution of Eq. (7-45b) is

$$E_x = E_0 e^{-\gamma z} = E_0 e^{-\alpha z} e^{-j\beta z}, \quad (7-46)$$

where both  $\alpha$  and  $\beta$  are positive quantities. The first factor,  $e^{-\alpha z}$ , decreases as  $z$  increases and thus is an attenuation factor, and  $\alpha$  is called an **attenuation constant**. The SI unit of the attenuation constant is neper per meter (Np/m).<sup>†</sup> The second factor,  $e^{-j\beta z}$ , is a phase factor;  $\beta$  is called a **phase constant** and is expressed in radians per meter (rad/m). The phase constant expresses the amount of phase shift that occurs as the wave travels one meter.

Attenuation constant  
and its SI unit

Phase constant and  
its SI unit

### EXERCISE 7.6

Assuming that the amplitude of the electric intensity of a plane wave propagating in a certain lossy medium is 1 (mV/m) at  $P_1$  and 0.8 (mV/m) at  $P_2$  50 (m) away, find

<sup>†</sup>Neper is a dimensionless quantity. If  $\alpha = 1$  (Np/m), then a unit wave amplitude decreases to a magnitude  $e^{-1} (= 0.368)$  as it travels a distance of 1 (m). An attenuation of 1 (Np/m) equals  $20 \log_{10} e = 8.69$  (dB/m).



- a) the total attenuation between points  $P_1$  and  $P_2$  both in nepers and in decibels,  
 b)  $\alpha$  in (Np/m) and in (dB/m).

ANS. (a) 0.223 (Np), 1.94 (dB), (b) 0.00446 (Np/m), 0.0388 (dB/m).

### 7-3.1 LOW-LOSS DIELECTRICS

A low-loss dielectric is a good but imperfect insulator with a nonzero equivalent conductivity, such that  $\epsilon'' \ll \epsilon'$  or  $\sigma/\omega\epsilon \ll 1$ . Under this condition,  $\gamma$  in Eq. (7-44) can be approximated by using the binomial expansion:

$$\gamma = \alpha + j\beta \cong j\omega\sqrt{\mu\epsilon'} \left[ 1 - j\frac{\epsilon''}{2\epsilon'} + \frac{1}{8}\left(\frac{\epsilon''}{\epsilon'}\right)^2 \right],$$

from which we obtain the attenuation constant

$$\alpha \approx \Re(\gamma) \cong \frac{\omega\epsilon''}{2} \sqrt{\frac{\mu}{\epsilon'}} \quad (\text{Np/m}) \quad (7-47)$$

and the phase constant

$$\beta = \Im(\gamma) \cong \omega\sqrt{\mu\epsilon'} \left[ 1 + \frac{1}{8}\left(\frac{\epsilon''}{\epsilon'}\right)^2 \right] \quad (\text{rad/m}). \quad (7-48)$$

It is seen from Eq. (7-47) that the attenuation constant of a low-loss dielectric is a positive quantity and is approximately directly proportional to the frequency. The phase constant in Eq. (7-48) deviates only very slightly from the value  $\omega\sqrt{\mu\epsilon'}$  for a perfect (lossless) dielectric.

The intrinsic impedance of a low-loss dielectric is a complex quantity.

$$\begin{aligned} \eta_c &= \sqrt{\frac{\mu}{\epsilon'}} \left( 1 - j\frac{\epsilon''}{\epsilon'} \right)^{-1/2} \\ &\cong \sqrt{\frac{\mu}{\epsilon'}} \left( 1 + j\frac{\epsilon''}{2\epsilon'} \right) \quad (\Omega). \end{aligned} \quad (7-49)$$

Since the intrinsic impedance is the ratio of  $E_x$  and  $H_y$  for a uniform plane wave, the electric and magnetic field intensities in a lossy dielectric are thus not in time phase, as they are in a lossless medium.

The phase velocity  $u_p$  is obtained from the ratio  $\omega/\beta$ . Using Eq. (7-48), we have

$$u_p = \frac{\omega}{\beta} \cong \frac{1}{\sqrt{\mu\epsilon'}} \left[ 1 - \frac{1}{8}\left(\frac{\epsilon''}{\epsilon'}\right)^2 \right] \quad (\text{m/s}), \quad (7-50)$$

which is slightly lower than its value when the medium is lossless.

Propagation  
constant of a lossy  
dielectric

Intrinsic impedance  
of a lossy dielectric

## REVIEW QUESTIONS

- Q.7-11** What makes the permittivity of a dielectric medium a complex quantity?
- Q.7-12** Define *loss tangent* of a medium.
- Q.7-13** What is the relation between *propagation constant* and wavenumber?
- Q.7-14** Define *attenuation constant* and *phase constant* of a wave propagating in a medium. What are their SI units?

## REMARKS

1. The electric and magnetic fields for uniform plane waves in a *lossy* medium are in space quadrature and have different time phases.
2. Both  $\alpha$  and  $\beta$  are real quantities and both are, in general, functions of frequency.
3. Attenuation of wave amplitude in nepers is the natural logarithm of the ratio of the amplitude at the beginning point to that at the end point.
4.  $1 \text{ (Np)} = 8.69 \text{ (dB)}$ .

## 7-3.2 GOOD CONDUCTORS

A good conductor is a medium for which  $\sigma/\omega\epsilon \gg 1$ . Under this condition it is convenient to use Eq. (7-43) and neglect 1 in comparison with  $\sigma/\omega\epsilon$ . We write

$$\gamma \cong j\omega\sqrt{\mu\epsilon}\sqrt{\frac{\sigma}{j\omega\epsilon}} = \sqrt{j}\sqrt{\omega\mu\sigma} = \frac{1+j}{\sqrt{2}}\sqrt{\omega\mu\sigma},$$

or

$$\gamma = \alpha + j\beta \cong (1+j)\sqrt{\pi f\mu\sigma}, \quad (7-51)$$

where we have used the relations

$$\sqrt{j} = (e^{j\pi/2})^{1/2} = e^{j\pi/4} = (1+j)/\sqrt{2}$$

and  $\omega = 2\pi f$ . Equation (7-51) indicates that  $\alpha$  and  $\beta$  for a good conductor are approximately equal and both increase as  $\sqrt{f}$  and  $\sqrt{\sigma}$ . For a good conductor,

$$\alpha = \beta = \sqrt{\pi f\mu\sigma}. \quad (7-52)$$

Propagation constant of a good conductor

Attenuation constant and phase constant of a good conductor are equal.

Relation between intrinsic impedance and attenuation constant of a good conductor

The intrinsic impedance of a good conductor is

$$\eta_c = \sqrt{\frac{\mu}{\epsilon_c}} \cong \sqrt{\frac{j\omega\mu}{\sigma}} = (1+j)\sqrt{\frac{\pi f\mu}{\sigma}} = (1+j)\frac{\alpha}{\sigma} \quad (\Omega), \quad (7-53)$$

which has a phase angle of  $45^\circ$ . Hence the magnetic field intensity lags behind the electric field intensity by  $45^\circ$ .

The phase velocity in a good conductor is

Phase velocity in a good conductor

$$u_p = \frac{\omega}{\beta} \cong \sqrt{\frac{2\omega}{\mu\sigma}} \quad (\text{m/s}), \quad (7-54)$$

which is proportional to  $\sqrt{f}$  and  $1/\sqrt{\sigma}$ . Consider copper as an example:

$$\sigma = 5.80 \times 10^7 \quad (\text{S/m}),$$

$$\mu = 4\pi \times 10^{-7} \quad (\text{H/m}),$$

$$u_p = 720 \text{ (m/s)} \quad \text{at} \quad 3 \text{ (MHz)},$$

which is many orders of magnitude slower than the velocity of light in air. The wavelength of a plane wave in a good conductor is

Wavelength in a good conductor

$$\lambda = \frac{2\pi}{\beta} = \frac{u_p}{f} = 2\sqrt{\frac{\pi}{f\mu\sigma}} \quad (\text{m}). \quad (7-55)$$

For copper at 3 (MHz),  $\lambda = 0.24 \text{ (mm)}$ . As a comparison, a 3-(MHz) electromagnetic wave in air has a wavelength of 100(m).

At very high frequencies the attenuation constant  $\alpha$  for a good conductor, as given by Eq. (7-52), tends to be very large. For copper at 3 (MHz),

$$\alpha = \sqrt{\pi(3 \times 10^6)(4\pi \times 10^{-7})(5.80 \times 10^7)} = 2.62 \times 10^4 \quad (\text{Np/m}).$$

Since the attenuation factor is  $e^{-\alpha z}$ , the amplitude of a wave will be attenuated by a factor of  $e^{-1} = 0.368$  when it travels a distance  $\delta = 1/\alpha$ . For copper at 3 (MHz) this distance is  $(1/2.62) \times 10^{-4} \text{ (m)}$ , or 0.038 (mm). At 10 (GHz) it is only 0.66 ( $\mu\text{m}$ )—a very small distance indeed. Thus a high-frequency electromagnetic wave is attenuated very rapidly as it propagates in a good conductor. The distance  $\delta$  through which the amplitude of a traveling plane wave decreases by a factor of  $e^{-1}$  or 0.368 is called the **skin depth** or the **depth of penetration** of a conductor:

Skin depth

Finding skin depth from conductivity and permeability of conductor and frequency

$$\delta = \frac{1}{\alpha} = \frac{1}{\sqrt{\pi f \mu \sigma}} \quad (\text{m}). \quad (7-56)$$

Since  $\alpha = \beta$  for a good conductor,  $\delta$  can also be written as

$$\delta = \frac{1}{\beta} = \frac{\lambda}{2\pi} \quad (\text{m}). \quad (7-57)$$

At microwave frequencies the skin depth or depth of penetration of a good conductor is so small that fields and currents can be considered as, for all practical purposes, confined in a very thin layer (that is, in the *skin*) of the conductor surface.

#### EXAMPLE 7-4

The electric field intensity of a linearly polarized uniform plane wave propagating in the  $+z$ -direction in seawater is  $\mathbf{E} = \mathbf{a}_x 100 \cos(10^7 \pi t)$  (V/m) at  $z = 0$ . The constitutive parameters of seawater are  $\epsilon_r = 72$ ,  $\mu_r = 1$ , and  $\sigma = 4$  (S/m).

- Determine the attenuation constant, phase constant, intrinsic impedance, phase velocity, wavelength, and skin depth.
- Find the distance at which the amplitude of  $\mathbf{E}$  is 1% of its value at  $z = 0$ .
- Write the expressions for  $\mathbf{E}(z, t)$  and  $\mathbf{H}(z, t)$  at  $z = 0.8$  (m) as functions of  $t$ .

#### SOLUTION

$$\omega = 10^7 \pi \quad (\text{rad/s}),$$

$$f = \frac{\omega}{2\pi} = 5 \times 10^6 \text{ (Hz)} = 5 \text{ (MHz)},$$

$$\frac{\sigma}{\omega\epsilon} = \frac{\sigma}{\omega\epsilon_0\epsilon_r} = \frac{4}{10^7 \pi \left( \frac{1}{36\pi} \times 10^{-9} \right) 72} = 200 \gg 1.$$

Hence we can use the formulas for good conductors.

- Attenuation constant:

$$\alpha = \sqrt{\pi f \mu \sigma} = \sqrt{5\pi 10^6 (4\pi 10^{-7}) 4} = 8.89 \quad (\text{Np/m}).$$

Phase constant:

$$\beta = \sqrt{\pi f \mu \sigma} = 8.89 \quad (\text{rad/m}).$$

Intrinsic impedance:

$$\begin{aligned} \eta_c &= (1 + j) \sqrt{\frac{\pi f \mu}{\sigma}} \\ &= (1 + j) \sqrt{\frac{\pi(5 \times 10^6)(4\pi \times 10^{-7})}{4}} = \pi e^{j\pi/4} \quad (\Omega). \end{aligned}$$



Phase velocity:

$$u_p = \frac{\omega}{\beta} = \frac{10^7 \pi}{8.89} = 3.53 \times 10^6 \quad (\text{m/s}).$$

Wavelength:

$$\lambda = \frac{2\pi}{\beta} = \frac{2\pi}{8.89} = 0.707 \quad (\text{m}).$$

Skin depth:

$$\delta = \frac{1}{\alpha} = \frac{1}{8.89} = 0.112 \quad (\text{m}).$$

- b) Distance  $z_1$  at which the amplitude of wave decreases to 1% of its value at  $z = 0$ :

$$e^{-\alpha z_1} = 0.01 \quad \text{or} \quad e^{\alpha z_1} = \frac{1}{0.01} = 100,$$

$$z_1 = \frac{1}{\alpha} \ln 100 = \frac{4.605}{8.89} = 0.518 \quad (\text{m}).$$

- c) In phasor notation,

$$\mathbf{E}(z) = \mathbf{a}_x 100 e^{-\alpha z} e^{-j\beta z}.$$

The instantaneous expression for  $\mathbf{E}$  is

$$\begin{aligned} \mathbf{E}(z, t) &= \Re[\mathbf{E}(z) e^{j\omega t}] \\ &= \Re[\mathbf{a}_x 100 e^{-\alpha z} e^{j(\omega t - \beta z)}] = \mathbf{a}_x 100 e^{-\alpha z} \cos(\omega t - \beta z). \end{aligned}$$

At  $z = 0.8$  (m) we have

$$\begin{aligned} \mathbf{E}(0.8, t) &= \mathbf{a}_x 100 e^{-0.8\alpha} \cos(10^7 \pi t - 0.8\beta) \\ &= \mathbf{a}_x 0.082 \cos(10^7 \pi t - 7.11) \\ &= \mathbf{a}_x 0.082 \cos(10^7 \pi t - 47.5^\circ)^{\dagger} \quad (\text{V/m}). \end{aligned}$$

We know that a uniform plane wave is a TEM wave with  $\mathbf{E} \perp \mathbf{H}$  and that both are normal to the direction of wave propagation  $\mathbf{a}_z$ . Thus  $\mathbf{H} = \mathbf{a}_y H_y$ . To find  $\mathbf{H}(z, t)$ , the instantaneous expression of  $\mathbf{H}$  as a function of  $t$ , we must not make the mistake of writing  $H_y(z, t) = E_x(z, t)/\eta_c$  because this would be mixing real time functions  $E_x(z, t)$  and  $H_z(z, t)$  with a complex quantity  $\eta_c$ . Phasor quantities  $E_x(z)$  and  $H_y(z)$  must be used. That is,

$$H_y(z) = \frac{E_x(z)}{\eta_c},$$

<sup>†</sup>7.11 (rad) =  $7.11 \times (180/\pi) = 407.4^\circ$ , which is equivalent to  $407.4^\circ - 360^\circ = 47.5^\circ$  in phase relations.

from which we obtain the relation between instantaneous quantities

$$H_y(z, t) = \Re \left[ \frac{E_x(z)}{\eta_c} e^{j\omega t} \right].$$

For the present problem we have, in phasors,

$$H_y(0.8) = \frac{100e^{-0.8\alpha}e^{-j0.8\beta}}{\pi e^{j\pi/4}} = \frac{0.082e^{-j7.11}}{\pi e^{j\pi/4}} = 0.026e^{-j1.61}.$$

Note that *both* angles must be in radians before combining. The instantaneous expression for  $\mathbf{H}$  at  $z = 0.8$  (m) is then

$$\begin{aligned} \mathbf{H}(0.8, t) &= \mathbf{a}_y 0.026 \cos(10^7 \pi t - 1.61) \\ &= \mathbf{a}_y 0.026 \cos(10^7 \pi t - 92.3^\circ) \quad (\text{A/m}). \end{aligned}$$

We can see that a 5-(MHz) plane wave attenuates very rapidly in seawater and becomes negligibly weak a very short distance from the source. (Field amplitudes at a depth of 0.8 (m) are reduced to  $0.082/100 = 0.00082$  times their value at the surface.) Even at very low frequencies, long-distance communication with a submerged submarine is very difficult.

#### EXERCISE 7.7

Determine the frequency at which the skin depth in seawater is ten meters. Calculate the corresponding wavelength in seawater and compare it to that in air.

ANS. 633 (Hz), 62.8 (m), 474 (km).

#### REVIEW QUESTIONS

**Q.7-15** What distinguishes a good conductor from a good insulator at a given frequency?

**Q.7-16** What is meant by the *skin depth* of a conductor?

#### REMARKS

1. The attenuation constant and the phase constant of a good conductor are numerically equal.
2. The intrinsic impedance of a good conductor has a phase angle of  $45^\circ$ .
3. The skin depth of a good conductor is numerically equal to the reciprocal of its attenuation constant, and is inversely proportional to  $\sqrt{f}$  and  $\sqrt{\sigma}$ .
4. The skin depth of good conductors is less than  $1 (\mu\text{m})$  at 10 (GHz).

## 7-4 GROUP VELOCITY

### Definition of phase velocity

In Eq. (7-10) we defined the phase velocity,  $u_p$ , of a single-frequency plane wave as the velocity of propagation of an equiphase wavefront. The relation between  $u_p$  and the phase constant,  $\beta$ , is

$$u_p = \frac{\omega}{\beta} \quad (\text{m/s}). \quad (7-58)$$

For plane waves in a lossless medium,  $\beta = k = \omega\sqrt{\mu\epsilon}$  is a linear function of  $\omega$ . As a consequence, the phase velocity  $u_p = 1/\sqrt{\mu\epsilon}$  is a constant that is independent of frequency. However, in some cases (such as wave propagation in a lossy dielectric, as discussed previously, or along a transmission line, or in a waveguide) the phase constant is not a linear function of  $\omega$ ; waves of different frequencies will propagate with different phase velocities. Because all information-bearing signals consist of a band of frequencies, waves of the component frequencies will travel with different phase velocities, causing a distortion in the signal wave shape. The signal “disperses.” The phenomenon of signal distortion caused by a dependence of the phase velocity on frequency is called **dispersion**. In view of Eqs. (7-50) and (7-39), we conclude that a lossy dielectric is a **dispersive medium**.

### Dispersion

An information-bearing signal normally has a small spread of frequencies (sidebands) around a high carrier frequency. Such a signal comprises a “group” of frequencies and forms a wave packet. A **group velocity** is the velocity of propagation of the wave-packet envelope (of a group of frequencies).

### Definition of Group velocity

Consider the simplest case of a wave packet that consists of two traveling waves having equal amplitude and slightly different angular frequencies  $\omega_0 + \Delta\omega$  and  $\omega_0 - \Delta\omega$  ( $\Delta\omega \ll \omega_0$ ). The phase constants, being functions of frequency, will also be slightly different. Let the phase constants corresponding to the two frequencies be  $\beta_0 + \Delta\beta$  and  $\beta_0 - \Delta\beta$ . We have

$$\begin{aligned} E(z, t) &= E_0 \cos [(\omega_0 + \Delta\omega)t - (\beta_0 + \Delta\beta)z] \\ &\quad + E_0 \cos [(\omega_0 - \Delta\omega)t - (\beta_0 - \Delta\beta)z] \\ &= 2E_0 \cos (t \Delta\omega - z \Delta\beta) \cos (\omega_0 t - \beta_0 z). \end{aligned} \quad (7-59)$$

Since  $\Delta\omega \ll \omega_0$ , the expression in Eq. (7-59) represents a rapidly oscillating wave having an angular frequency  $\omega_0$  and an amplitude that varies slowly with an angular frequency  $\Delta\omega$ . This wave pattern is depicted in Fig. 7-6.

The wave inside the envelope propagates with a phase velocity found by setting  $\omega_0 t - \beta_0 z = \text{Constant}$ :

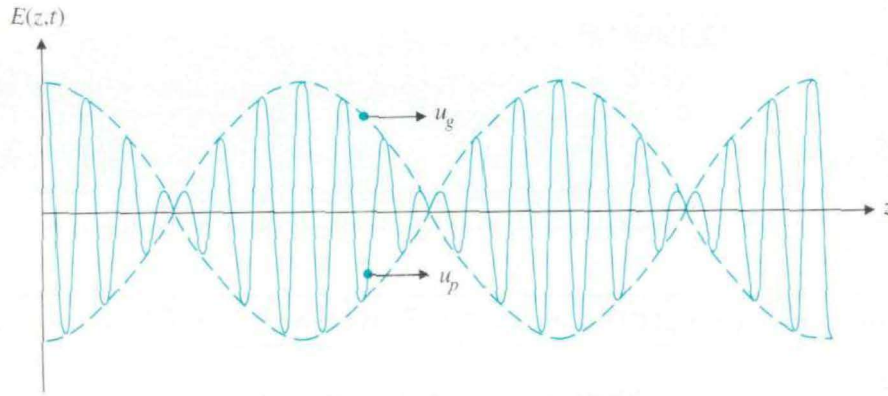


FIGURE 7-6 Sum of two time-harmonic traveling waves of equal amplitude and slightly different frequencies at a given  $t$ .

$$u_p = \frac{dz}{dt} = \frac{\omega_0}{\beta_0}.$$

The velocity of the envelope (the **group velocity**  $u_g$ ) can be determined by setting the argument of the first cosine factor in Eq. (7-58) equal to a constant:

$$t \Delta\omega - z \Delta\beta = \text{Constant},$$

from which we obtain

$$u_g = \frac{dz}{dt} = \frac{\Delta\omega}{\Delta\beta} = \frac{1}{\Delta\beta/\Delta\omega}.$$

In the limit that  $\Delta\omega \rightarrow 0$ , we have the formula for computing the group velocity in a dispersive medium:

Formula for group  
velocity in dispersive  
media

$$u_g = \frac{1}{d\beta/d\omega} \quad (\text{m/s}).$$

(7-60)

This is the velocity of a point on the envelope of the wave packet, as shown in Fig. 7-6, and is identified as the velocity of the narrow-band signal. Group velocity in a dispersive medium may be higher or lower than phase velocity. A medium is said to exhibit normal dispersion if  $u_g < u_p$ , and to exhibit anomalous dispersion if  $u_g > u_p$ . When  $u_g = u_p$ , there is no dispersion.

### REVIEW QUESTIONS

**Q.7-17** What is meant by the *dispersion* of a signal? Give an example of a dispersive medium.

**Q.7-18** Define *group velocity*. In what ways is group velocity different from phase velocity?



**REMARKS**

1. Information-bearing signals propagate with no distortion only in nondispersive media.
2. A medium is nondispersive if  $\beta$  is a linear function of (directly proportional to)  $\omega$ .

**7-5 FLOW OF ELECTROMAGNETIC POWER AND THE POYNTING VECTOR**

Electromagnetic waves carry with them electromagnetic power. Energy is transported through space to distant receiving points by electromagnetic waves. We will now derive a relation between the rate of such energy transfer and the electric and magnetic field intensities associated with a traveling electromagnetic wave.

We begin with the curl equations:

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}, \quad (6-45a)(7-61)$$

$$\nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t}. \quad (6-45b)(7-62)$$

The following identity of vector operations can be verified in a straightforward manner by using Cartesian coordinates:

$$\nabla \cdot (\mathbf{E} \times \mathbf{H}) = \mathbf{H} \cdot (\nabla \times \mathbf{E}) - \mathbf{E} \cdot (\nabla \times \mathbf{H}). \quad (7-63)$$

Substitution of Eqs. (7-61) and (7-62) in Eq. (7-63) yields

$$\nabla \cdot (\mathbf{E} \times \mathbf{H}) = -\mathbf{H} \cdot \frac{\partial \mathbf{B}}{\partial t} - \mathbf{E} \cdot \frac{\partial \mathbf{D}}{\partial t} - \mathbf{E} \cdot \mathbf{J}. \quad (7-64)$$

In a simple medium, whose constitutive parameters  $\epsilon$ ,  $\mu$ , and  $\sigma$  do not change with time, we have

$$\mathbf{H} \cdot \frac{\partial \mathbf{B}}{\partial t} = \mathbf{H} \cdot \frac{\partial(\mu \mathbf{H})}{\partial t} = \frac{1}{2} \frac{\partial(\mu \mathbf{H} \cdot \mathbf{H})}{\partial t} = \frac{\partial}{\partial t} \left( \frac{1}{2} \mu H^2 \right),$$

$$\mathbf{E} \cdot \frac{\partial \mathbf{D}}{\partial t} = \mathbf{E} \cdot \frac{\partial(\epsilon \mathbf{E})}{\partial t} = \frac{1}{2} \frac{\partial(\epsilon \mathbf{E} \cdot \mathbf{E})}{\partial t} = \frac{\partial}{\partial t} \left( \frac{1}{2} \epsilon E^2 \right),$$

$$\mathbf{E} \cdot \mathbf{J} = \mathbf{E} \cdot (\sigma \mathbf{E}) = \sigma E^2.$$

Equation (7-64) can then be written as

$$\nabla \cdot (\mathbf{E} \times \mathbf{H}) = -\frac{\partial}{\partial t} \left( \frac{1}{2} \epsilon E^2 + \frac{1}{2} \mu H^2 \right) - \sigma E^2, \quad (7-65)$$

which is a point-function relationship. An integral form of Eq. (7-65) is obtained by integrating both sides over the volume of concern:

$$\oint_S (\mathbf{E} \times \mathbf{H}) \cdot d\mathbf{s} = -\frac{\partial}{\partial t} \int_V \left( \frac{1}{2} \epsilon E^2 + \frac{1}{2} \mu H^2 \right) dv - \int_V \sigma E^2 dv, \quad (7-66)$$

where the divergence theorem has been applied to convert the volume integral of  $\nabla \cdot (\mathbf{E} \times \mathbf{H})$  to the closed surface integral of  $(\mathbf{E} \times \mathbf{H})$ .

We recognize that the first and second terms on the right side of Eq. (7-66) represent the time-rate of change of the energy stored in the electric and magnetic fields, respectively. [Compare with Eqs. (3-106) and (5-106).] The last term is the ohmic power dissipated in the volume as a result of the flow of conduction current density  $\sigma \mathbf{E}$  in the presence of the electric field  $\mathbf{E}$ . Hence we may interpret the right side of Eq. (7-66) as the *rate of decrease* of the electric and magnetic energies stored, subtracted by the ohmic power dissipated as heat in the volume  $V$ . To be consistent with the law of conservation of energy, this must equal the power (rate of energy) *leaving* the volume through its surface. Thus the quantity  $(\mathbf{E} \times \mathbf{H})$  is a vector representing the power flow per unit area. Define

$$\mathcal{P} = \mathbf{E} \times \mathbf{H} \quad (\text{W/m}^2). \quad (7-67)$$

Quantity  $\mathcal{P}$  is known as the **Poynting vector**, which is a power density vector associated with an electromagnetic field. The assertion that the surface integral of  $\mathcal{P}$  over a closed surface, as given by the left side of Eq. (7-66), equals the power leaving the enclosed volume is referred to as **Poynting's theorem**. This assertion is not limited to plane waves.

Equation (7-66) may be written in another form:

$$-\oint_S \mathcal{P} \cdot d\mathbf{s} = \frac{\partial}{\partial t} \int_V (w_e + w_m) dv + \int_V p_\sigma dv, \quad (7-68)$$

where

$$w_e = \frac{1}{2} \epsilon E^2 = \frac{1}{2} \epsilon \mathbf{E} \cdot \mathbf{E}^* = \text{Electric energy density}, \quad (7-69)$$

$$w_m = \frac{1}{2} \mu H^2 = \frac{1}{2} \mu \mathbf{H} \cdot \mathbf{H}^* = \text{Magnetic energy density}, \quad (7-70)$$

$$p_\sigma = \sigma E^2 = J^2 / \sigma = \sigma \mathbf{E} \cdot \mathbf{E}^* = \mathbf{J} \cdot \mathbf{J}^* / \sigma = \text{Ohmic power density}. \quad (7-71)$$

In words, Eq. (7-68) states that the total power flowing *into* a closed surface at any instant equals the sum of the rates of increase of the stored electric and magnetic energies and the ohmic power dissipated within the enclosed volume. An asterisk on a quantity denotes the complex conjugate of that quantity.

Definition of  
Poynting vector

Poynting's theorem

**EXAMPLE 7-5**

Find the Poynting vector on the surface of a long, straight conducting wire (of radius  $b$  and conductivity  $\sigma$ ) that carries a direct current  $I$ . Verify Poynting's theorem.

**SOLUTION**

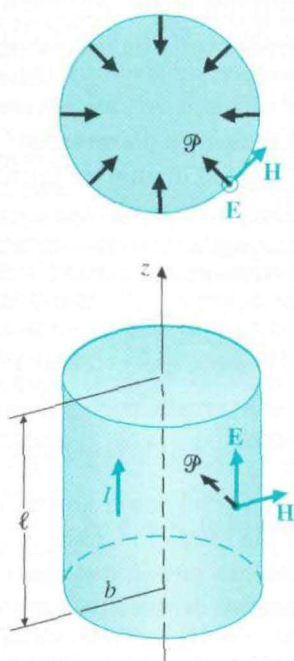
Since we have a d-c situation, the current in the wire is uniformly distributed over its cross-sectional area. Let us assume that the axis of the wire coincides with the  $z$ -axis. Figure 7-7 shows a segment of length  $\ell$  of the long wire. We have

$$\mathbf{J} = \mathbf{a}_z \frac{I}{\pi b^2}$$

and

$$\mathbf{E} = \frac{\mathbf{J}}{\sigma} = \mathbf{a}_z \frac{I}{\sigma \pi b^2}.$$

FIGURE 7-7 Illustrating Poynting's theorem (Example 7-5).



On the surface of the wire,

$$\mathbf{H} = \mathbf{a}_\phi \frac{I}{2\pi b}.$$

Thus the Poynting vector at the surface of the wire is

$$\begin{aligned}\mathcal{P} &= \mathbf{E} \times \mathbf{H} = (\mathbf{a}_z \times \mathbf{a}_\phi) \frac{I^2}{2\sigma\pi^2 b^3} \\ &= -\mathbf{a}_r \frac{I^2}{2\sigma\pi^2 b^3},\end{aligned}$$

which is directed everywhere into the wire surface.

To verify Poynting's theorem, we integrate  $\mathcal{P}$  over the wall of the wire segment in Fig. 7-7:

$$\begin{aligned}-\oint_S \mathcal{P} \cdot d\mathbf{s} &= -\oint_S \mathcal{P} \cdot \mathbf{a}_r ds = \left( \frac{I^2}{2\sigma\pi^2 b^3} \right) 2\pi b \ell \\ &= I^2 \left( \frac{\ell}{\sigma\pi b^2} \right) = I^2 R,\end{aligned}$$

where the formula for the resistance of a straight wire in Eq. (4-16),  $R = \ell/\sigma S$ , has been used. The above result affirms that the negative surface integral of the Poynting vector is exactly equal to the  $I^2 R$  ohmic power loss in the conducting wire. Hence Poynting's theorem is verified.

### 7-5.1 INSTANTANEOUS AND AVERAGE POWER DENSITIES

In dealing with time-harmonic electromagnetic waves we have found it convenient to use phasor notation. The instantaneous value of a quantity is then the real part of the product of the phasor quantity and  $e^{j\omega t}$  when  $\cos \omega t$  is used as the reference. For example, for the phasor

$$\mathbf{E}(z) = \mathbf{a}_x E_x(z) = \mathbf{a}_x E_0 e^{-(\alpha + j\beta)z}, \quad (7-72)$$

the instantaneous expression is

$$\begin{aligned}\mathbf{E}(z, t) &= \Re_e[\mathbf{E}(z)e^{j\omega t}] = \mathbf{a}_x E_0 e^{-\alpha z} \Re_e[e^{j(\omega t - \beta z)}] \\ &= \mathbf{a}_x E_0 e^{-\alpha z} \cos(\omega t - \beta z).\end{aligned} \quad (7-73)$$

For a uniform plane wave propagating in a lossy medium in the  $+z$ -direction, the associated magnetic field intensity phasor is

$$\mathbf{H}(z) = \mathbf{a}_y H_y(z) = \mathbf{a}_y \frac{E_0}{|\eta_c|} e^{-\alpha z} e^{-j(\beta z + \theta_\eta)}, \quad (7-74)$$

where  $\theta_\eta$  is the phase angle of the intrinsic impedance  $\eta_c = |\eta_c|e^{j\theta_\eta}$  of the

Writing the  
instantaneous  
expression from a  
phasor



medium. The corresponding instantaneous expression for  $\mathbf{H}(z)$  is

$$\mathbf{H}(z, t) = \Re \epsilon [\mathbf{H}(z)e^{j\omega t}] = \mathbf{a}_y \frac{E_0}{|\eta_c|} e^{-\alpha z} \cos(\omega t - \beta z - \theta_\eta). \quad (7-75)$$

The instantaneous expression for the Poynting vector or power density vector, from Eqs. (7-72) and (7-74), is

$$\begin{aligned} \mathcal{P}(z, t) &= \mathbf{E}(z, t) \times \mathbf{H}(z, t) = \Re \epsilon [\mathbf{E}(z)e^{j\omega t}] \times \Re \epsilon [\mathbf{H}(z)e^{j\omega t}] \\ &= \mathbf{a}_z \frac{E_0^2}{|\eta_c|} e^{-2\alpha z} \cos(\omega t - \beta z) \cos(\omega t - \beta z - \theta_\eta) \\ &= \mathbf{a}_z \frac{E_0^2}{2|\eta_c|} e^{-2\alpha z} [\cos \theta_\eta + \cos(2\omega t - 2\beta z - \theta_\eta)]. \end{aligned} \quad (7-76)^*$$

As far as the power transmitted by an electromagnetic wave is concerned, its average value is a more significant quantity than its instantaneous value. From Eq. (7-76), we obtain the time-average Poynting vector,  $\mathcal{P}_{av}(z)$ :

Average power density transmitted by a uniform plane wave in  $z$ -direction

$$\mathcal{P}_{av}(z) = \frac{1}{T} \int_0^T \mathcal{P}(z, t) dt = \mathbf{a}_z \frac{E_0^2}{2|\eta_c|} e^{-2\alpha z} \cos \theta_\eta \quad (\text{W/m}^2), \quad (7-77)$$

where  $T = 2\pi/\omega$  is the time period of the wave. The second term on the right side of Eq. (7-76) is a cosine function of a double frequency whose average is zero over a fundamental period. For wave propagation in lossless media,  $\eta_c \rightarrow \eta$  is real,  $\sigma = 0$ , and  $\theta_\eta = 0$ , Eq. (7-77) reduces to

$$\mathcal{P}_{av}(z) = \mathbf{a}_z \frac{E_0^2}{2\eta} \quad (\text{W/m}^2). \quad (7-78)$$

In the general case we may not be dealing with a wave propagating in the  $z$ -direction. We write

General formula for average power density in a propagating wave

$$\mathcal{P}_{av} = \frac{1}{2} \Re \epsilon (\mathbf{E} \times \mathbf{H}^*) \quad (\text{W/m}^2), \quad (7-79)$$

which is a general formula for computing the average power density in a propagating wave.

### EXERCISE 7.8

Substitute the phasor expressions of  $\mathbf{E}(z)$  and  $\mathbf{H}(z)$  given by Eqs. (7-72) and (7-74) in Eq. (7-79) to verify  $\mathcal{P}_{av}$  obtained in Eq. (7-77).

\*Here we have used the trigonometric identity  $\cos \theta_1 \cos \theta_2 = \frac{1}{2} [\cos(\theta_1 - \theta_2) + \cos(\theta_1 + \theta_2)]$ .

**EXAMPLE 7-6**

The phasor expressions of the far field at a distance  $R$  from a short vertical current element  $I d\ell$  located in free space at the origin of a spherical coordinate system are

$$\mathbf{E}(R, \theta) = \mathbf{a}_\theta E_\theta(R, \theta) = \mathbf{a}_\theta \left( j \frac{60\pi I d\ell}{\lambda R} \sin \theta \right) e^{-j\beta R} \quad (\text{V/m}) \quad (7-80)$$

and

$$\mathbf{H}(R, \theta) = \mathbf{a}_\phi \frac{E_\theta(R, \theta)}{\eta_0} = \mathbf{a}_\phi \left( j \frac{I d\ell}{2\lambda R} \sin \theta \right) e^{-j\beta R} \quad (\text{A/m}), \quad (7-81)$$

where  $\lambda = 2\pi/\beta$  is the wavelength.

- Write the expression for instantaneous Poynting vector.
- Find the total average power radiated by the current element.

**SOLUTION**

- We note that  $E_\theta/H_\phi = \eta_0 = 120\pi(\Omega)$ . The instantaneous Poynting vector is

$$\begin{aligned} \mathcal{P}(R, \theta; t) &= \Re e[\mathbf{E}(R, \theta)e^{j\omega t}] \times \Re e[\mathbf{H}(R, \theta)e^{j\omega t}] \\ &= (\mathbf{a}_\theta \times \mathbf{a}_\phi) 30\pi \left( \frac{I d\ell}{\lambda R} \right)^2 \sin^2 \theta \sin^2(\omega t - \beta R) \\ &= \mathbf{a}_R 15\pi \left( \frac{I d\ell}{\lambda R} \right)^2 \sin^2 \theta [1 - \cos 2(\omega t - \beta R)] \quad (\text{W/m}^2). \end{aligned}$$

- The average power density vector is, from Eq. (7-79),

$$\mathcal{P}_{\text{av}}(R, \theta) = \mathbf{a}_R 15\pi \left( \frac{I d\ell}{\lambda R} \right)^2 \sin^2 \theta, \quad (7-82)$$

which is seen to equal the time-average value of  $\mathcal{P}(R, \theta; t)$  given in part (a) of this solution. The total average power radiated is obtained by integrating  $\mathcal{P}_{\text{av}}(R, \theta)$  over the surface of the sphere of radius  $R$ :

$$\begin{aligned} \text{Total } P_{\text{av}} &= \oint_S \mathcal{P}_{\text{av}}(R, \theta) \cdot d\mathbf{s} \\ &= \int_0^{2\pi} \int_0^\pi \left[ 15\pi \left( \frac{I d\ell}{\lambda R} \right)^2 \sin^2 \theta \right] R^2 \sin \theta d\theta d\phi \\ &= 40\pi^2 \left( \frac{d\ell}{\lambda} \right)^2 I^2 \quad (\text{W}), \end{aligned} \quad (7-83)$$

where  $I$  is the amplitude ( $\sqrt{2}$  times the effective value) of the sinusoidal current in  $d\ell$ .